기출문제집

10/15 에 교수님이 알려 준 11문제 순서로, 같거나 비슷한 기출 (그래프신경망 가을 중간/기말 2023~2025, 같은 교수님의 그래프마이닝 봄) 의 문제와 공식 해설을 그대로 모았어요. 문제마다 이번 시험의 어느 소문항과 같은지, 해설에서 고칠 점을 적었어요. 먼저 문제만 보고 풀어 본 뒤 해설을 보세요.

01 메시지 패싱: 인접행렬 정규화, GCN 한 층, 순열 등변성 — 기출 6개
2023 가을 중간 1번, 2025 가을 중간 3번, 2024 가을 중간 1번, 2023 가을 중간 4번, 2023 가을 중간 3번, 2024 가을 기말 1번

02 WL relabeling 과 GIN — 기출 5개
2026 봄 그래프마이닝 기말 3번, 2023 봄 그래프마이닝 기말 7번, 2024 봄 그래프마이닝 중간 11번, 2024 가을 기말 5번, 2025 가을 기말 1번

03 ESAN (bag of subgraphs) 과 1-WL 넘는 방법 비교 — 기출 6개
2024 가을 기말 6번, 2025 가을 기말 2번, 2024 가을 기말 7번, 2025 가을 기말 5번, 2024 가을 기말 10번, 2025 가을 기말 3번

04 계산 그래프와 계산 부담, full batch — 기출 2개
2024 가을 기말 2번, 2025 가을 중간 5번 (a)

05 Cluster-GCN 과 그 문제점 — 기출 5개
2025 가을 중간 5번, 2024 가을 중간 3번, 2023 가을 기말 2번, 2025 가을 중간 6번, 2024 가을 중간 4번

06 과평활, 과압축, under-reaching 과 저역통과 필터 — 기출 3개
2024 가을 기말 3번, 2025 가을 중간 7번, 2025 가을 중간 9번

07 GCNII: initial residual, identity mapping — 기출 4개
2025 가을 중간 8번, 2024 가을 중간 6번, 2024 가을 기말 3번, 2024 가을 중간 7번

08 동질성 지표 (homophily metrics) — 기출 없음
6주차는 올해 새 내용. 사이트 카드의 직접 만든 예제로 연습

09 H2GCN — 기출 없음
6주차는 올해 새 내용. 사이트 카드의 직접 만든 예제로 연습

10 GAT 어텐션과 GAT 대 GATv2 — 기출 4개
2025 가을 중간 10번, 2025 가을 중간 11번, 2024 가을 중간 8번, 2024 가을 기말 4번

11 HAN: 동작과 어텐션을 두 번 하는 이유 — 기출 3개
2025 가을 중간 12번, 2024 가을 중간 9번, 2025 가을 기말 8번

01 메시지 패싱: 인접행렬 정규화, GCN 한 층, 순열 등변성

2주차, 목요일

(a) 인접행렬 정규화: $D^{-1}A$ (행 정규화) 또는 $\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}$ (대칭), $\tilde{A} = A + I$

(b) GCN 한 층 손계산: $H^{(1)} = \sigma(\hat{A}H^{(0)}W)$

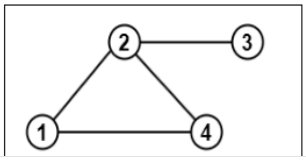
(c) 순열 등변성: 노드 번호를 바꿔도 왜 출력이 (같이 바뀔 뿐) 같은가. $Pf(A, X) = f(PAP^T, PX)$

2023 가을 중간 1번 정규화 인접행렬 계산 + 2-step 전이확률

(a) 와 같은 꼴

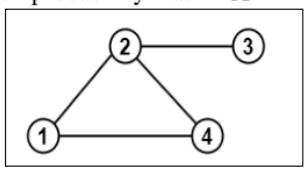
문제

1. Consider an undirected graph G of four nodes given in the following figure. The normalized adjacency matrix is defined as: $\tilde{A} = D^{-1} \cdot A$, where A is the adjacency matrix of G and D is the node degree matrix.
   a) Calculate the matrix $\tilde{A}$
   b) Calculate the 2-step transition probability matrix $\tilde{A}^2$



공식 해설

1. Consider an undirected graph G of four nodes given in the following figure. The normalized adjacency matrix is defined as: $\tilde{A} = D^{-1} \cdot A$, where A is the adjacency matrix of G and D is the node degree matrix.
   a) Calculate the matrix $\tilde{A}$
   b) Calculate the 2-step transition probability matrix $\tilde{A}^2$



Solutions:

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} \qquad D = \begin{bmatrix} 2 \\ 3 \\ 1 \\ 2 \end{bmatrix}$$

$$\tilde{A} = \frac{A}{D} = \begin{bmatrix} 0 & .5 & 0 & .5 \\ .33 & 0 & .33 & .33 \\ 0 & 1 & 0 & 0 \\ .5 & .5 & 0 & 0 \end{bmatrix}$$

$$\tilde{A}^2 = \tilde{A} \times \tilde{A} = \begin{bmatrix} 0 & .5 & 0 & .5 \\ .33 & 0 & .33 & .33 \\ 0 & 1 & 0 & 0 \\ .5 & .5 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & .5 & 0 & .5 \\ .33 & 0 & .33 & .33 \\ 0 & 1 & 0 & 0 \\ .5 & .5 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0.42 & 0.25 & 0.17 & 0.17 \\ 0.17 & 0.67 & 0 & 0.17 \\ 0.33 & 0 & 0.33 & 0.33 \\ 0.17 & 0.25 & 0.17 & 0.42 \end{bmatrix}$$

2025 가을 중간 3번 (a) GCN 층에서 정보가 퍼지는 법 (b) σ(AH) 계산 **(b) 와 거의 그대로**

문제

3. (GCN)
   a. Explain how information is propagated in a GCN layer.
   b. Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix} \qquad H^{(0)} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 2 \\ 2 & 2 \\ 2 & 0 \end{bmatrix}$$

Assume that the hidden layer of an GCN model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma\left(A \cdot H^{(k-1)}\right),$$

where $H^{(k)}$ denotes the output at layer k, σ is a ReLU function $\mathrm{ReLU}(x) = \max(0, x)$.
Calculate the output of the GCN model at layer $k = 1$.

공식 해설

3. (GCN) (10p)
   a. (3p) Explain how information is propagated in a GCN layer.
   Each node aggregates and averages features from its neighbors (using the normalized adjacency matrix) and then applies a linear transformation and activation, updating its representation based on local graph structure
   b. (7p) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

2

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix} \qquad H^{(0)} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 2 \\ 2 & 2 \\ 2 & 0 \end{bmatrix}$$

Assume that the hidden layer of an GCN model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma\left(A \cdot H^{(k-1)}\right),$$

where $H^{(k)}$ denotes the output at layer k, σ is a ReLU function $\mathrm{ReLU}(x) = \max(0, x)$.
Calculate the output of the GCN model at layer $k = 1$.

Ans:

$$H^1 = \sigma(AH^0) = \mathrm{ReLU}\left(\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 2 \\ 2 & 2 \\ 2 & 0 \end{bmatrix}\right) = \mathrm{ReLU}\left(\begin{bmatrix} 2 & 2 \\ 3 & 2 \\ 1 & 0 \\ 3 & 5 \\ 2 & 3 \end{bmatrix}\right) = \begin{bmatrix} 2 & 2 \\ 3 & 2 \\ 1 & 0 \\ 3 & 5 \\ 2 & 3 \end{bmatrix}$$

2024 가을 중간 1번 σ(AH) 로 GCN k=1 계산 (음수가 섞임) (b) 와 같은 꼴

문제

1. (10pt) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix} \qquad H^{(0)} = \begin{bmatrix} -2 & 1 \\ -3 & -3 \\ -1 & 4 \\ 2 & 2 \\ 4 & 0 \end{bmatrix}$$

Assume that the hidden layer of an GCN model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma\left(A \cdot H^{(k-1)}\right),$$

where $H^{(k)}$ denotes the output at layer k, σ is a ReLU function $\mathrm{ReLU}(x) = \max(0, x)$. Calculate the output of the GCN model at layer $k = 1$.

공식 해설

1. (10pt) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix} \qquad H^{(0)} = \begin{bmatrix} -2 & 1 \\ -3 & -3 \\ -1 & 4 \\ 2 & 2 \\ 4 & 0 \end{bmatrix}$$

Assume that the hidden layer of an GCN model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma\left(A \cdot H^{(k-1)}\right),$$

where $H^{(k)}$ denotes the output at layer k, σ is a ReLU function $\mathrm{ReLU}(x) = \max(0, x)$. Calculate the output of the GCN model at layer $k = 1$.

Ans:

$$H^1 = \sigma(AH^0) = \mathrm{ReLU}\left(\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} -2 & 1 \\ -3 & -3 \\ -1 & 4 \\ 2 & 2 \\ 4 & 0 \end{bmatrix}\right) = \mathrm{ReLU}\left(\begin{bmatrix} 2 & 2 \\ 4 & 0 \\ -1 & -1 \\ -1 & 7 \\ -4 & 4 \end{bmatrix}\right)$$

$$= \begin{bmatrix} 2 & 2 \\ 4 & 0 \\ 0 & 0 \\ 0 & 7 \\ 0 & 4 \end{bmatrix}$$

2023 가을 중간 4번 σ(AH) 로 GCN k=1 계산 (b) 와 같은 꼴

문제

4. Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix} \qquad H^{(0)} = \begin{bmatrix} 0 & 0 \\ 1 & -1 \\ 2 & -2 \\ 3 & -3 \\ 4 & -4 \\ 5 & -5 \end{bmatrix}$$

Assume that the hidden layer of an GCN model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma\left(A \cdot H^{(k-1)}\right),$$

where $H^{(k)}$ denotes the output at layer k, σ is a ReLU function $\mathrm{ReLU}(x) = \max(0, x)$. Calculate the output of the GCN model at layer $k = 1$.

2

4. Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A=\begin{bmatrix}0&1&1&1&0&0\\1&0&1&0&0&0\\1&1&0&0&0&0\\1&0&0&0&1&1\\0&0&0&1&0&1\\0&0&0&1&1&0\end{bmatrix} \qquad H^{(0)}=\begin{bmatrix}0&0\\1&-1\\2&-2\\3&-3\\4&-4\\5&-5\end{bmatrix}$$

Assume that the hidden layer of an GCN model of all nodes at layer (k) can be calculated as:

$$H^{(k)}=\sigma\left(A\cdot H^{(k-1)}\right),$$

where $H^{(k)}$ denotes the output at layer k, σ is a ReLU function $\mathrm{ReLU}(x)=\max(0,x)$. Calculate the output of the GCN model at layer $k=1$.

Solutions:

$$H^{1}=\sigma\left(AH^{0}\right)=\mathrm{ReLU}\left(\begin{bmatrix}0&1&1&1&0&0\\1&0&1&0&0&0\\1&1&0&0&0&0\\1&0&0&0&1&1\\0&0&0&1&0&1\\0&0&0&1&1&0\end{bmatrix}\times\begin{bmatrix}0&0\\1&-1\\2&-2\\3&-3\\4&-4\\5&-5\end{bmatrix}\right)=\mathrm{ReLU}\left(\begin{bmatrix}6&-6\\2&-2\\1&-1\\9&-9\\8&-8\\7&-7\end{bmatrix}\right)=\begin{bmatrix}6&0\\2&0\\1&0\\9&0\\8&0\\7&0\end{bmatrix}$$

3

2023 가을 중간 3번 이웃 평균으로 k=1, k=2 전파 (b) 와 비슷 (한 층을 손으로)

문제

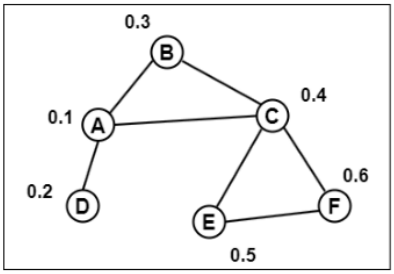
3. Consider an undirected graph G of six nodes A, B, C, D, E, and F given in the following figure. Each node has initial features that are the numbers standing next to it (i.e., the initial feature of node 'A' is $X_A^{(0)} = 0.1$). Assume that the feature of a node i at layer k can be updated as:

$$X_i^{(k)} = F\left(X_i^{(k-1)}, N_i^{(k-1)}\right),$$

where $F(\bullet)$ returns a numeric value that is the mean (average) of its arguments, $N_i^{(k-1)}$ is the set of features of the neighbours of node i at layer $(k-1)$.

a) Calculate the feature of each node at $k = 1$.

b) Calculate the feature of each node at $k = 2$.



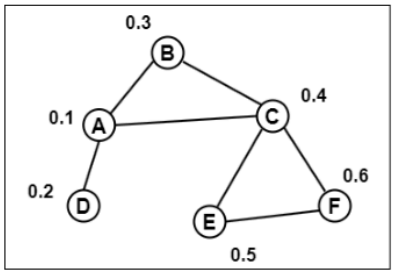
1

3. Consider an undirected graph G of six nodes A, B, C, D, E, and F given in the following figure. Each node has initial features that are the numbers standing next to it (i.e., the initial feature of node 'A' is $X_A^{(0)} = 0.1$). Assume that the feature of a node i at layer k can be updated as:

$$X_i^{(k)} = F\left(X_i^{(k-1)}, N_i^{(k-1)}\right),$$

where $F(\bullet)$ returns a numeric value that is the mean (average) of its arguments, $N_i^{(k-1)}$ is the set of features of the neighbours of node i at layer $(k-1)$.

a) Calculate the feature of each node at $k = 1$.
b) Calculate the feature of each node at $k = 2$.



Solutions:

$$X_A^1 = mean\left(X_A^0 + X_B^0 + X_C^0 + X_D^0\right) = \frac{0.1+0.2+0.3+0.4}{4} = 0.25$$

$$X_B^1 = mean\left(X_A^0 + X_B^0 + X_C^0\right) = (0.1+0.3+0.4)/3 = 0.27$$

$$X_C^1 = (0.1+0.3+0.4+0.5+0.6)/5 = 0.38$$

$$X_D^1 = (0.1+0.2)/2 = 0.15$$

$$X_E^1 = (0.4+0.5+0.6)/3 = 0.5$$

$$X_F^1 = (0.4+0.5+0.6)/3 = 0.5$$

2

$$X_B^2 = mean\left(X_A^1 + X_B^1 + X_C^1\right) = (0.25+0.27+0.38)/3 = 0.3$$

$$X_C^2 = (0.25+0.27+0.38+0.5+0.5)/5 = 0.38$$

$$X_D^2 = (0.25+0.15)/2 = 0.2$$

$$X_E^2 = (0.38+0.5+0.5)/3 = 0.46$$

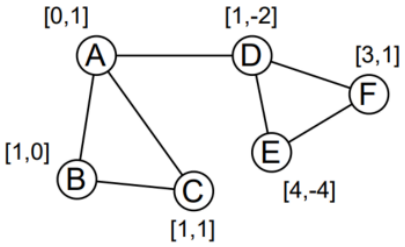
$$X_F^2 = (0.38+0.5+0.5)/3 = 0.46$$

고칠 점: 해설 그림 두 장 사이에서 X_A(2) = 0.26 줄이 잘려 빠졌어요 (원문에는 있음)

2024 가을 기말 1번 GCN 계산 + GraphSAGE 와 비교한 GCN 의 한계 (b) 와 비슷

문제

1. (10p) Consider an undirected graph G of six nodes A, B, C, D, E, and F given in the following figure. Each node has initial features that are the numbers standing next to it.



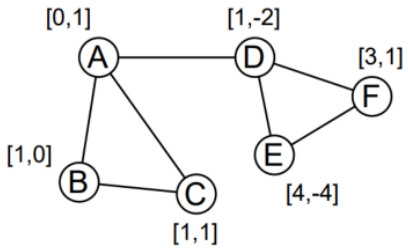
Assume that the hidden layer of an GCN model of all nodes at layer k can be calculated as:

$$H^{(k)} = \sigma\left(A \cdot H^{(k-1)}\right),$$

where $H^{(k)}$ denotes the output at layer k, σ is a ReLU function $\mathrm{ReLU}(x) = \max(0, x)$.
A is the adjacency matrix.

a) Calculate the output of the GCN model at layer $k = 1$.
b) What are the limitations of the GCN model compared to GraphSAGE?

1. (10p) Consider an undirected graph G of six nodes A, B, C, D, E, and F given in the following figure. Each node has initial features that are the numbers standing next to it, as follows:



Assume that the hidden layer of an GCN model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma\left(A \cdot H^{(k-1)}\right),$$

where $H^{(k)}$ denotes the output at layer k, σ is a ReLU function $\mathrm{ReLU}(x) = \max(0, x)$.
A is the adjacency matrix.

a. Calculate the output of the GCN model at layer $k = 1$.
b. What are the limitations of the GCN model compared to GraphSage?

Solutions:

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

$$H^1 = \sigma\left(AH^0\right) = \mathrm{ReLU}\left(\begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \\ 1 & -2 \\ 4 & -4 \\ 3 & 1 \end{bmatrix}\right) = \mathrm{ReLU}\left(\begin{bmatrix} 3 & -1 \\ 1 & 2 \\ 1 & 1 \\ 7 & -2 \\ 4 & -1 \\ 5 & -6 \end{bmatrix}\right) = \begin{bmatrix} 3 & 0 \\ 1 & 2 \\ 1 & 1 \\ 7 & 0 \\ 4 & 0 \\ 5 & 0 \end{bmatrix}$$

The limitations of GCN model compared to GraphSage:

- Scalability: GCNs require the entire graph adjacency matrix and feature matrix during training. This can lead to high memory usage and computational inefficiency due to matrix multiplications involving the entire graph.
- Transductive learning: GCNs assume a static, fixed graph structure during training. Any changes to the graph (e.g., adding nodes or edges) require retraining the model or re-computing the embeddings.

1

02 WL relabeling 과 GIN

3주차, 목요일

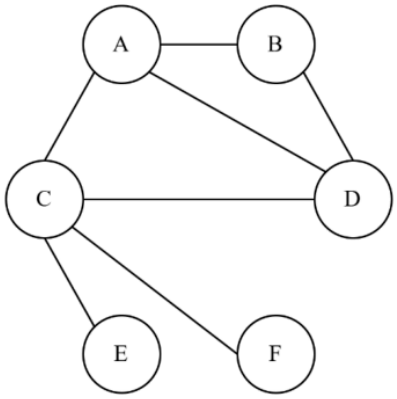
(a) WL relabeling (1-WL color refinement) 을 그래프에 손으로: 처음 색 1, 이웃 색 멀티셋, 새 색 붙이기, 반복. 두 그래프 구별
(b) GIN 은 왜 보통 GNN (평균, 최댓값 집계) 보다 표현력이 강한가: 단사 집계 = 합 + MLP, 1-WL 과 같은 힘

2026 봄 그래프마이닝 기말 3번 1-WL 한 번 color refinement (처음 색 1, 이웃 색 멀티셋, 새 색) + 1-WL 이 못 가리는 그래프 예 **(a) 와 거의 그대로**

문제

Question 3: (Weiseiler-Lehman Kernel) (10 points)

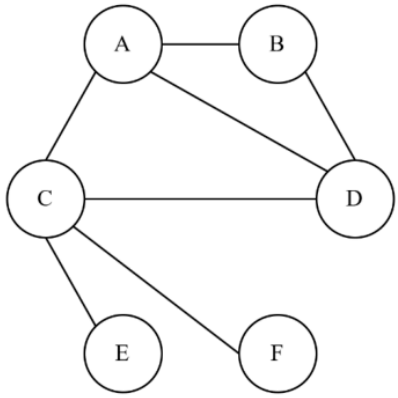
a) Perform one iteration (color refinement) of the 1-WL test for the given graph. Initialize all nodes with the color "1". Show the multi-set of neighborhood colors for each node, the aggregated string, and assign new hashed colors. (5 points)



b) Provide an example of a simple graph structure that the 1-WL test is provably incapable of detecting or counting. (5 points)

Question 3: (Weiseiler-Lehman Kernel) (10 points)

a) Perform one iteration (color refinement) of the 1-WL test for the given graph. Initialize all nodes with the color "1". Show the multi-set of neighborhood colors for each node, the aggregated string, and assign new hashed colors. (5 points)



b) Provide an example of a simple graph structure that the 1-WL test is provably incapable of detecting or counting. (5 points)

Solution:

a) We first determine the adjacency list for each node to identify their neighborhoods $\mathcal{N}(v)$:
$\mathcal{N}(A) = \{B, C, D\}$
$\mathcal{N}(B) = \{A, D\}$
$\mathcal{N}(C) = \{A, D, E, F\}$
$\mathcal{N}(D) = \{A, B, C\}$
$\mathcal{N}(E) = \{C\}$
$\mathcal{N}(F) = \{C\}$
Initial State, all nodes are initialized with the same feature/color:

$$c^{(0)}(v) = 1 \quad \forall v \in \{A, B, C, D, E, F\}$$

First Iteration
Step 1: Compute neighborhood color multisets

Node	Degree	Neighborhood color multiset
A	3	$\{1, 1, 1\}$
B	2	$\{1, 1\}$
C	4	$\{1, 1, 1, 1\}$
D	3	$\{1, 1, 1\}$
E	1	$\{1\}$
F	1	$\{1\}$

Step 2: Form aggregated strings

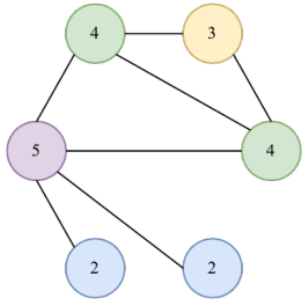
Node	Aggregated String
A	$(1,\{1,1,1\})$
B	$(1,\{1,1\})$
C	$(1,\{1,1,1,1\})$
D	$(1,\{1,1,1\})$
E	$(1,\{1\})$
F	$(1,\{1\})$

Step 3: Hash aggregated strings into new colors

Aggregated String Input	Injective Hash Output
$(1,\{1\})$	2
$(1,\{1,1\})$	3
$(1,\{1,1,1\})$	4
$(1,\{1,1,1,1\})$	5

Thus

Node	New Hashed Color
A	4
B	3
C	5
D	4
E	2
F	2



b) A classic and provable example is a triangle (a cycle of 3 nodes).
Specifically, the 1-WL test is completely incapable of distinguishing between:

- A single, continuous hexagon (a cycle of 6 nodes, C_6).
- Two disconnected triangles (two sets of 3-node cycles, $2 \times C_3$).

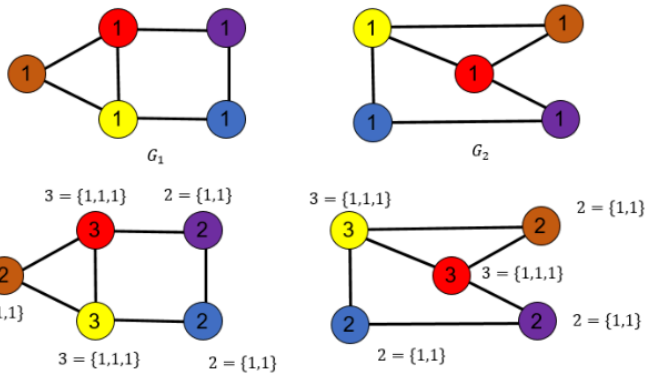
2023 봄 그래프마이닝 기말 7번 WL relabeling 후 두 그래프가 동형인지 판정 (해설에 문제 포함) (a) 와 같은 꼴 공식 해설

7. Consider two undirected graphs in the following figure. (10pt)
   a) Conduct Weisfeiler-Lehman (WL) relabeling process with the maximum degree, 3. Then, using the Weisfeiler-Lehman isomorphism testing, determine whether two graphs are isomorphic or not?

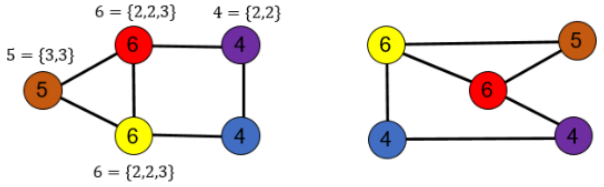
   Two graphs are isomorphic:

   Step 1: set node label =1 for all nodes

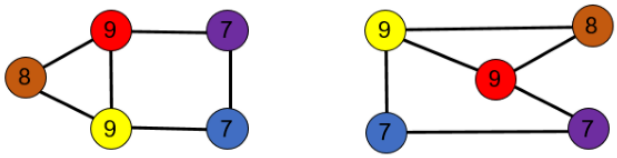
4



Step 2:



Step 3:

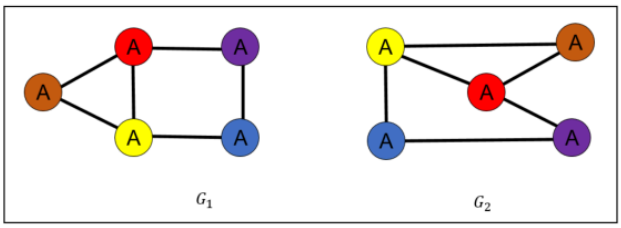


b) Make feature vectors for the graphs based on frequency of node degrees.

In G_1 , there are two nodes with degree 3, three nodes with degree 2, $f(G_1)=(2,3)$

In G_2 , there are two nodes with degree 3, three nodes with degree 2, $f(G_2)=(2,3)$

c) Make feature vectors for the graphs based on frequency of the WL subgraphs.



Based on the WL relabeling process in (a), the number of WL subgraphs in G_1 is as follows:

The number of label “1”: 5

The number of label “2”: 3

5

The number of label “3”: 2

The number of label “4”: 2

The number of label “5”: 1

The number of label “6”: 2

The number of label “7”: 2

The number of label “8”: 1

The number of label “9”: 2

Therefore, the feature vector for G_1 is: (5,3,2,2,1,2,2,1,2)

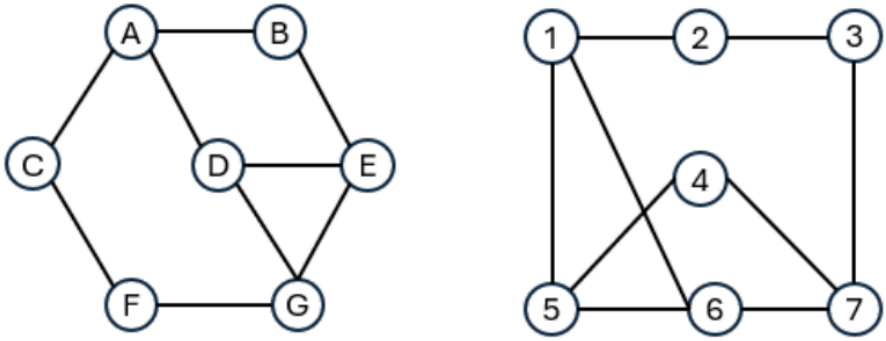
Similarly, the feature vector for G_2 is: (5,3,2,2,1,2,2,1,2)

고칠 점: 해설 2라운드의 4 = {2,2} 는 {2,3} 의 오타. 결론 isomorphic 은 1-WL 만으로는 '못 가린다' 까지가 맞는 말 (두 그래프는 실제로 동형)

2024 봄 그래프마이닝 중간 11번 WL relabeling + WL 부분그래프 빈도 벡터 + 코사인 유사도 (a) 와 비슷

문제

11. Consider two undirected graphs in the following figure: G_1 on the left and G_2 on the right. (10pt)
    a. Conduct Weisfeiler-Lehman (WL) relabeling process with the maximum degree 3.
    b. Make feature vectors for the graphs based on frequency of the WL subgraphs. Then calculate the similarity of graph G_1 and G_2 using Cosine Similarity equation (1).

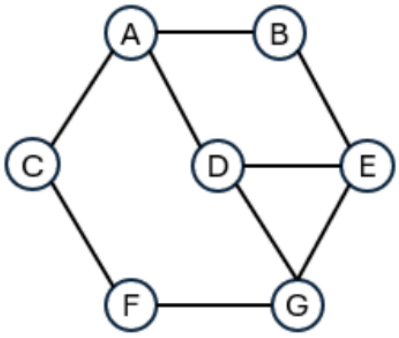


G_1 G_2

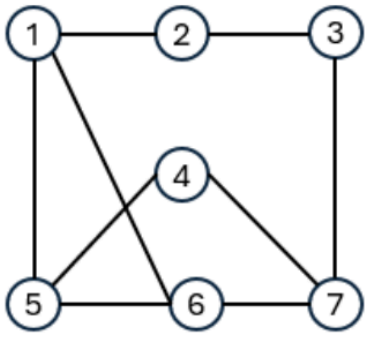
Cosine Similarity Equation (1): $cosine(WL_{G_1}, WL_{G_2}) = \frac{WL_{G_1}.WL_{G_2}}{\|WL_{G_1}\|\|WL_{G_2}\|}$.

where WL_{G_1} and WL_{G_2} is feature vectors of WL subgraph G_1 and G_2. “.” denotes the dot product and “|| ||” denotes the Euclidean norm.

11. Consider two undirected graphs in the following figure: G_1 on the left and G_2 on the right. (10pt)
    a. Conduct Weisfeiler-Lehman (WL) relabeling process with the maximum degree 3.
    b. Make feature vectors for the graphs based on frequency of the WL subgraphs in question a. Then calculate the similarity of graph G_1 and G_2 using equation (1).



G_1



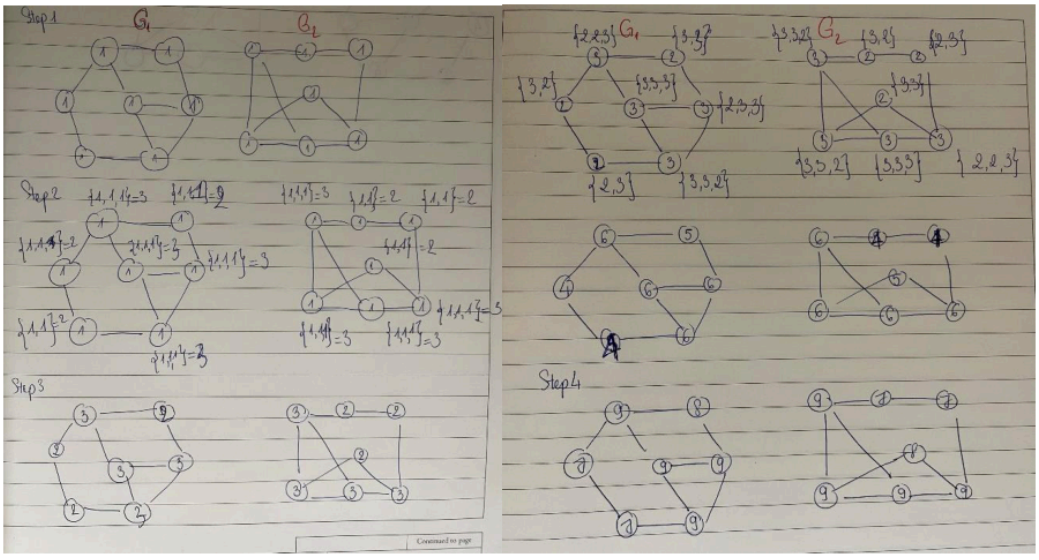
G_2

Cosine Similarity equation 1: $cosine(WL_{G_1}, WL_{G_2}) = \frac{WL_{G_1}.WL_{G_2}}{\|WL_{G_1}\|\|WL_{G_2}\|}$.

where WL_{G_1} and WL_{G_2} is feature vectors of WL subgraph G_1 and G_2. "." denotes the dot product and "|| ||" denotes the Euclidean norm.

Ans:

a.



b. Based on the WL relabeling process in (a), the number of WL subgraphs in G_1 is as follows:

The number of label "1": 7

The number of label “2”: 3

The number of label “3”: 4

The number of label “4”: 2

The number of label “5”: 1

The number of label “6”: 4

The number of label “7”: 2

The number of label “8”: 1

The number of label “9”: 4

Therefore, the feature vector for G_1 is: (7,3,4,2,1,4,2,1,4)

Similarly, the feature vector for G_2 is: (7,3,4,2,1,4,2,1,4)

The similarity of G_1 and G_2 is

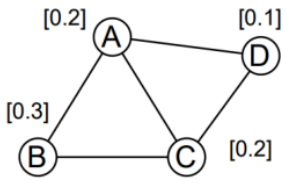
$$\frac{7*7+3*3+4*4+2*2+1*1+4*4+2*2+1*1+4*4}{\sqrt{7^2+3^2+4^2+2^2+1^1+4^2+2^2+1^2+4^2}.\sqrt{7^2+3^2+4^2+2^2+1^1+4^2+2^2+1^2+4^2}}=1$$

고칠 점: 손글씨 해설이 2라운드에서 서로 다른 멀티셋 ({2,2,3} 과 {3,3,3}) 에 같은 번호 6 을 줬어요. 제대로 해도 히스토그램이 끝까지 같아 코사인 1 결론은 같음

2024 가을 기말 5번 GIN 계산 (ε = 0.5) + Why GIN achieves 1-WL? **(b) 와 거의 그대로**

문제

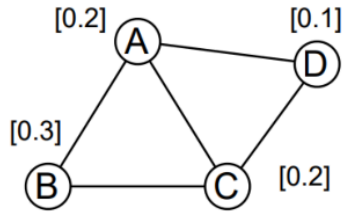
5. (10p) Consider an undirected graph G of four nodes A, B, C, and D given in the following figure. Each node has an initial feature that is the number standing next to it. According to GIN model, the parameter is a fixed scalar $\varepsilon = 0.5$, the feature of a node i at layer k can be updated as:



$$h_i^{(k)} = (1+\varepsilon)\cdot h_i^{(k-1)} + \sum_{j\in N(i)} h_j^{(k-1)}$$

a) Calculate the feature of each node at $k = 1$.
b) What are the reasons why GIN model achieves 1-d Weisfeiler-Lehman?

5. (10p) Consider an undirected graph G of four nodes A, B, C, and D given in the following figure. Each node has initial features that are the numbers standing next to it. According to GIN model, the parameter is a fixed scalar $\varepsilon = 0.5$, the feature of a node i at layer k can be updated as:



$$h_i^{(k)} = (1+\varepsilon)\cdot h_i^{(k-1)} + \sum_{j\in N(i)} h_j^{(k-1)}$$

a. Calculate the feature of each node at $k = 1$.

b. What are the reasons why GIN model achieves 1-d Weisfeiler-Lehman?

SOLUTIONS:

5

$$h_i^{(k)} = (1+\varepsilon)\cdot h_i^{(k-1)} + \sum_{j\in N(i)} h_j^{(k-1)}$$

$$h_A^{(1)} = (1+\varepsilon)\cdot h_A^{(0)} + \left(h_B^{(0)} + h_C^{(0)} + h_D^{(0)}\right) = (1+0.5)*0.2 + (0.3+0.2+0.1) = 0.3+0.6 = 0.9$$

$$h_B^{(1)} = (1+\varepsilon)\cdot h_B^{(0)} + \left(h_A^{(0)} + h_C^{(0)}\right) = (1+0.5)*0.3 + (0.2+0.2) = 0.45+0.4 = 0.85$$

$$h_C^{(1)} = (1+\varepsilon)\cdot h_C^{(0)} + \left(h_A^{(0)} + h_B^{(0)} + h_D^{(0)}\right) = (1+0.5)*0.2 + (0.2+0.3+0.1) = 0.3+0.6 = 0.9$$

$$h_D^{(1)} = (1+\varepsilon)\cdot h_D^{(0)} + \left(h_A^{(0)} + h_C^{(0)}\right) = (1+0.5)*0.1 + (0.2+0.2) = 0.15+0.4 = 0.55$$

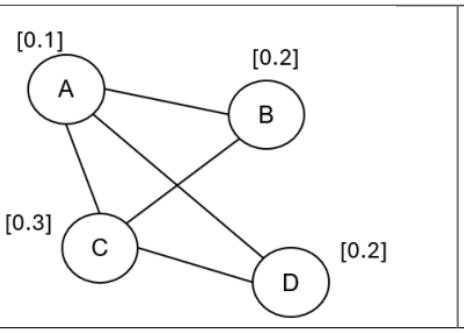
GIN uses a sum aggregation function to combine the features of neighboring nodes. Unlike mean or max-pooling used in other GNNs, the sum operation ensures that the aggregation function is injective (i.e., distinct input features produce distinct outputs).

The combination of the sum operator with an injective MLP ensures that GIN can distinguish graphs as powerful as the 1-WL testing.

2025 가을 기말 1번 GIN 계산 + Why GIN achieves 1-WL? (그래프만 다름) **(b) 와 거의 그대로**

문제

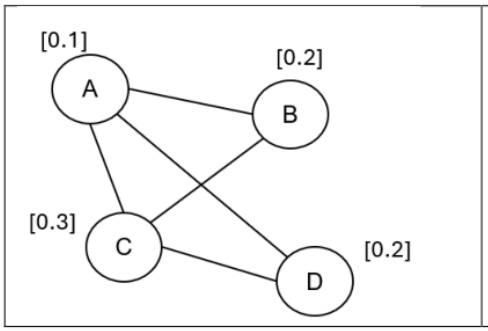
1. (GIN)(10p) Consider an undirected graph G of four nodes A, B, C, and D given in the following figure. Each node has an initial feature that is the number standing next to it. According to GIN model, the parameter is a fixed scalar $\varepsilon = 0.5$, the feature of a node i at layer k can be updated as:



$$h_i^{(k)} = (1+\varepsilon)\cdot h_i^{(k-1)} + \sum_{j\in N(i)} h_j^{(k-1)}$$

a) Calculate the feature of each node at $k = 1$.
b) What are the reasons why GIN model achieves 1-d Weisfeiler-Lehman?

1. (GIN)(10p) Consider an undirected graph G of four nodes A, B, C, and D given in the following figure. Each node has an initial feature that is the number standing next to it. According to GIN model, the parameter is a fixed scalar $\varepsilon = 0.5$, the feature of a node i at layer k can be updated as:



$$h_i^{(k)} = (1+\varepsilon)\cdot h_i^{(k-1)} + \sum_{j \in N(i)} h_j^{(k-1)}$$

a) Calculate the feature of each node at $k = 1$.

b) What are the reasons why GIN model achieves 1-d Weisfeiler-Lehman?

SOLUTIONS:

a) $h_i^{(k)} = (1+\varepsilon).h_i^{(k-1)} + \sum_{j \epsilon N(i)} h_j^{(k-1)}$

$h_A^{(1)} = (1+\varepsilon).h_A^{(0)} + \left(h_B^{(0)} + h_C^{(0)} + h_D^{(0)}\right) = (1+0.5)*0.1 + (0.2+0.3+0.2)$

$= 0.85$

$h_B^{(1)} = (1+\varepsilon).h_B^{(0)} + \left(h_A^{(0)} + h_C^{(0)}\right) = (1+0.5)*0.2 + (0.1+0.3) = 0.7$

$h_C^{(1)} = (1+\varepsilon).h_C^{(0)} + \left(h_A^{(0)} + h_B^{(0)} + h_D^{(0)}\right) = (1+0.5)*0.3 + (0.1+0.2+0.2)$

$= 0.95$

$h_D^{(1)} = (1+\varepsilon).h_D^{(0)} + \left(h_A^{(0)} + h_C^{(0)}\right) = (1+0.5)*0.2 + (0.1+0.3) = 0.7$

b) GIN uses a sum aggregation function to combine the features of neighboring nodes. Unlike mean or max pooling used in other GNNs, the sum operation ensures that the aggregation function is injective (i.e., distinct input features produce distinct outputs). The combination of the sum operator with an injective MLP ensures that GIN can distinguish graphs as powerful as the 1-WL testing.

03 ESAN (bag of subgraphs) 과 1-WL 넘는 방법 비교

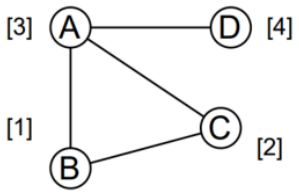
3주차, 목요일

(a) ESAN: 그래프를 부분그래프 묶음(bag, 노드/엣지 하나씩 지우기 등 정책 π)으로 바꿔 각각 GNN, 모아서 판정. 왜 1-WL 보다 강한가
(b) 1-WL 을 넘는 방법 비교 토론: 구조 특징 주입 (GSN, substructure count) / bag of subgraphs (ESAN) / 고차 GNN (k-GNN) 의 장단점

2024 가을 기말 6번 GNN-AK: k홉 부분그래프 계산 + 왜 k홉 부분그래프가 이웃 집계보다 표현력이 큰가 (a) 와 같은 갈래 (부분그래프)

문제

6. (10p) Consider an undirected graph G of four nodes A, B, C, and D given in the following figure. According to GNN-AK model with a GCN model and MEAN aggregation function, the feature of a node v at k-th layer can be updated as:



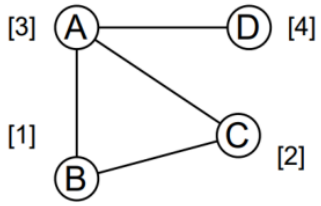
$$h_v^{(l+1)} = AGG^{(l)}(G[N_k(v)]), l = 0,1,\ldots,L-1$$
$$h_v = MEAN\left(h_u^{(L)} | u \in G[N_k(v)]\right)$$

where $G[N_k(v)]$ is a subgraph rooted at node v with k-hop distance, including v; and AGG(.) is a SUM function.

a) Calculate the feature of each node at $k = 1$ and $l = 0$.
b) Why considering a set of k-hop subgraphs will make the model more expressive than only neighbor aggregation?

2

6. (10p) Consider an undirected graph G of four nodes A, B, C, and D given in the following figure. According to GNN-AK model (From start to subgraph) with a GCN model and MEAN aggregation function, the feature of a node i at layer k can be updated as:



$$h_v^{(l+1)} = AGG^{(l)}\left(G\left[N_k(v)\right]\right), l = 0,1,...,L-1$$

$$h_v = MEAN\left(h_u^{(L)} \mid u \in G\left[N_k(v)\right]\right)$$

where $G\left[N_k(v)\right]$ is the subgraph rooted at node v with k-hop distance including v, and AGG(.) is a SUM function.

a. Calculate the feature of each node at $k = 1$ and $l = 0$.
b. Why considering a set of k-hop subgraphs will make the model more expressive than only neighbor aggregation?

SOLUTIONS:

$$h_A^{(1)} = AGG^{(0)}(G[A,B,C,D]) = SUM(1+2+3+4) = 10$$
$$h_B^{(1)} = SUM(1,2,3) = 6$$
$$h_C^{(1)} = SUM(1,2,3) = 6$$
$$h_D^{(1)} = SUM(3,4) = 7$$
$$h_A = MEAN\left(h_u^{(L)} | u \in G[N_k(v)]\right) = MEAN(10+6+6+7) = 7.25$$
$$---$$
$$h_B^{(1)} = AGG^{(0)}(G[A,B,C]) = SUM(1,2,3) = 6$$
$$h_C^{(1)} = SUM(1,2,3) = 6$$
$$h_A^{(1)} = SUM(1,2,3) = 6$$
$$h_B = MEAN\left(h_u^{(L)} | u \in G[N_k(v)]\right) = MEAN(6+6+6) = 6$$
$$---$$
$$h_C^{(1)} = AGG^{(0)}(G[A,B,C]) = SUM(1,2,3) = 6$$

6

$$h_B^{(1)} = SUM(1,2,3) = 6$$
$$h_A^{(1)} = SUM(1,2,3) = 6$$
$$h_C = MEAN\left(h_u^{(L)} | u \in G[N_k(v)]\right) = MEAN(6+6+6) = 6$$
$$---$$
$$h_D^{(1)} = AGG^{(0)}(G[A,D]) = SUM(3,4) = 7$$
$$h_A^{(1)} = SUM(3,4) = 7$$
$$h_D = MEAN\left(h_u^{(L)} | u \in G[N_1(D)]\right) = MEAN(7,7) = 7$$

A k-hop subgraph retains the entire subgraph's structural information (e.g., edge arrangements, motifs, and surrounding connectivity) within the k-hop distance, providing a more complete local representation. The higher-order structures, such as cycles, or cliques, are inherently captured separately within k-hop subgraphs, allowing models to leverage complex structural patterns.

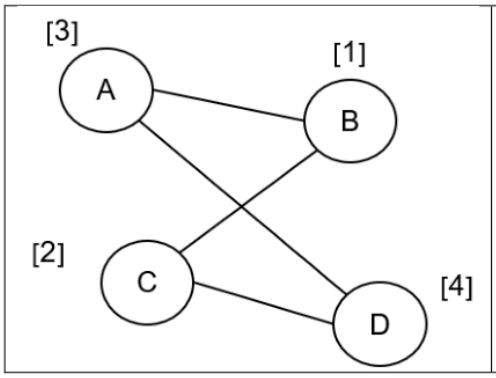
Neighbor aggregation cannot distinguish nodes in different structural roles or higher-order graph features, as it focuses only on immediate neighbors. K-hop subgraphs, on the other hand, explicitly incorporate these relationships.

고칠 점: 이 해설은 h_u 를 v 의 부분그래프 안에서 다시 계산해 (7.25, 6, 6, 7). 2025 해설처럼 식을 글자 그대로 (전체 그래프에서) 풀면 (7.25, 7.33, 7.33, 8.5). 시험에선 답과 어느 읽기인지 함께

2025 가을 기말 2번 GNN-AK (2024 기말 6번과 같은 문제, 머리에 '(K-GNN)' 이라고 적힘) (a) 와 같은 갈래

문제

2. (K-GNN)(10p) Consider an undirected graph G of four nodes A, B, C, and D given in the following figure. According to GNN-AK model with a GCN model and MEAN aggregation function, the feature of a node v at k-th layer can be updated as:



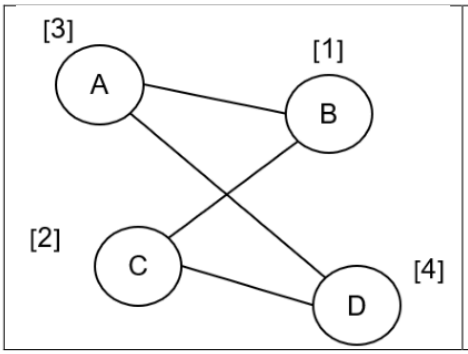
$h_v^{(l+1)} = AGG^{(l)}(G[N_k(v)]), l = 0,1,\ldots,L-1$
$h_v = MEAN(h_u^{(L)} | u \in G[N_k(v)])$
where $G[N_k(v)]$ is a subgraph rooted at node v with k-hop distance, including v; and AGG(.) is a SUM function.

a) Calculate the feature of each node at $k = 1$ and $l = 0$.
b) Why considering a set of k-hop subgraphs will make the model more expressive than only neighbor aggregation?

공식 해설

2. (K-GNN)(10p) Consider an undirected graph G of four nodes A, B, C, and D given in the following figure. According to GNN-AK model with a GCN model and MEAN aggregation function, the feature of a node v can be updated as:

1



$h_v^{(l+1)} = AGG^{(l)}(G[N_k(v)]), l = 0,1,\ldots,L-1$
$h_v = MEAN\left(h_u^{(L)} | u \in G[N_k(v)]\right)$
where $G\left[N_k(v)\right]$ is a subgraph rooted at node v with k-hop distance, including v; and AGG(.) is a SUM function.

a) Calculate the feature of each node at $k = 1$ and $l = 0$.
b) Why considering a set of k-hop subgraphs will make the model more expressive than only neighbor aggregation?

SOLUTIONS:

a) $h_A^{(1)} = AGG^{(0)}(G[A,B,D]) = SUM(1,3,4) = 8$
$h_B^{(1)} = SUM(1,2,3) = 6$
$h_D^{(1)} = SUM(2,3,4) = 9$
$h_A = MEAN\left(h_u^{(L)} | u \in G[N_k(v)]\right) = MEAN(8,6,9) = \frac{23}{3}$

$h_B^{(1)} = SUM(1,2,3) = 6$
$h_A^{(1)} = SUM(1,3,4) = 8$
$h_C^{(1)} = SUM(1,2,4) = 7$
$h_B = MEAN(6,8,7) = 7$

$h_C^{(1)} = SUM(1,2,4) = 7$
$h_B^{(1)} = SUM(1,2,3) = 6$
$h_D^{(1)} = SUM(2,3,4) = 9$
$h_C = MEAN(6,7,9) = 7.3$

$h_D^{(1)} = SUM(2,3,4) = 9$
$h_A^{(1)} = SUM(1,3,4) = 8$
$h_C^{(1)} = SUM(1,2,4) = 7$
$h_D = MEAN(9,8,7) = 8$

b) A k-hop subgraph retains the entire subgraph's structural information (e.g., edge arrangements, motifs, and surrounding connectivity) within the k-hop distance, providing a more complete local representation. The higher-order structures, such as cycles, or

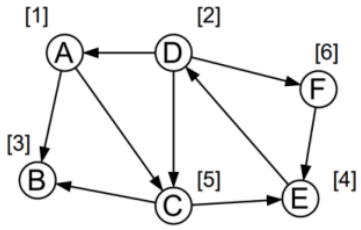
cliques, are inherently captured separately within k-hop subgraphs, allowing models to leverage complex structural patterns.
Neighbor aggregation cannot distinguish nodes in different structural roles or higher-order graph features, as it focuses only on immediate neighbors. K-hop subgraphs, on the other hand, explicitly incorporate these relationships.

고칠 점: 이 해설은 식을 글자 그대로 읽어 h_u 를 전체 그래프에서 구함. 22/3 을 7.3 으로 적음. 2024 해설과 읽는 법이 다름

2024 가을 기말 7번 Graphormer: 구조 정보를 넣으면 왜 강해지나, 어떤 구조 정보를 (b) 의 '구조 특징 주입' 부분

문제

7. (10p) Given a directed graph G of six nodes A, B, C, D, E, and F given in the following figure. Each node has an initial feature that is the number standing next to it.

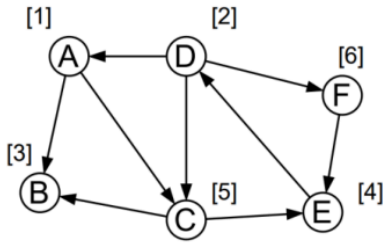


a) According to the Graphormer model, calculate the input node feature with its in-degree and out-degree. The definition of a Lookup table for node degree is as: T = {"0": 0.0, "1": 0.1, “2”: 0.2, “3”: 0.3, “4”: 0.4}.

b) Why can injecting the structural information into graphs make the model more powerful in capturing the graph structures? What kind of structural information needs to be considered?

공식 해설

7. (10p) Given a directed graph G of six nodes A, B, C, D, E, and F given in the following figure



a. According to the Graphormer model, calculate the initial node features of each node with Centrality Encoding. The definition of a Lookup table for node degree is as follows: T = {"0": 0.0, "1": 0.1, “2”: 0.2, “3”: 0.3, “4”: 0.4}.

b. Why encode the structural information of a graph in the model to make the model more powerful in capturing the graph structures? What kind of structural information needs to be considered?

SOLUTIONS

$$h_A^0 = h_A + d_A^- + d_A^+ = 1 + T[1] + T[2] = 1 + 0.1 + 0.2 = 1.3$$
$$h_B^0 = h_B + d_B^- + d_B^+ = 3 + T[0] + T[2] = 3 + 0.2 = 3.2$$
$$h_C^0 = 5 + T[2] + T[2] = 5 + 0.2 + 0.2 = 5.4$$
$$h_D^0 = 2 + T[3] + T[1] = 2 + 0.3 + 0.1 = 2.4$$
$$h_E^0 = 4 + T[1] + T[2] = 4 + 0.1 + 0.2 = 4.3$$
$$h_F^0 = 6 + T[1] + T[1] = 6.2$$

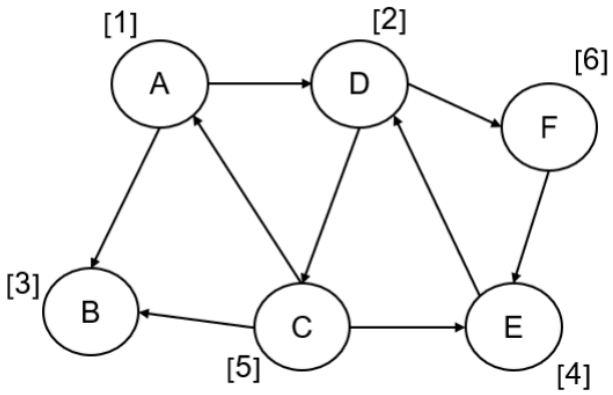
7

Structural information helps the model distinguish similar graphs when considering only node-level features or simple neighbor aggregation, allowing the model to identify non-isomorphic graphs. Encoding structural properties like degree, subgraph structures, or motifs allows the model to identify such distinctions and increase the power of graph isomorphism tests as the Weisfeiler-Lehman (WL) test.

There are several kinds of structural information, which can be integrated into the model: Laplacian Eigenvectors, SPD, Random Walk, and structural identity.

2025 가을 기말 5번 Graphormer (2024 기말 7번과 같은 문제) (b) 의 '구조 특징 주입' 부분 문제

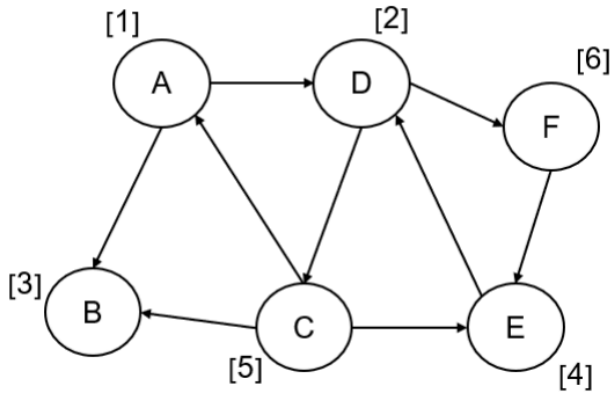
5. (Graphformer)(10p) Given a directed graph G of six nodes A, B, C, D, E, and F given in the following figure. Each node has an initial feature that is the number standing next to it.



a) According to the Graphormer model, calculate the input node feature with its in-degree and out-degree. The definition of a Lookup table for node degree is as: T = {"0": 0.0, "1": 0.1, “2”: 0.2, “3”: 0.3, “4”: 0.4}.

b) Why can injecting the structural information into graphs make the model more powerful in capturing the graph structures? What kind of structural information needs to be considered?

5. (Graphformer)(10p) Given a directed graph G of six nodes A, B, C, D, E, and F given in the following figure. Each node has an initial feature that is the number standing next to it.



a) According to the Graphormer model, calculate the input node feature with its in-degree and out-degree. The definition of a Lookup table for node degree is as: T = {"0": 0.0, "1": 0.1, "2": 0.2, "3": 0.3, "4": 0.4}.

b) Why can injecting the structural information into graphs make the model more powerful in capturing the graph structures? What kind of structural information needs to be considered?

SOLUTIONS:

a)
$$h_A^0 = h_A + d_A^- + d_A^+ = 1 + T[2] + T[1] = 1 + 0.2 + 0.1 = 1.3$$
$$h_B^0 = h_B + d_B^- + d_B^+ = 3 + T[0] + T[2] = 3 + 0 + 0.2 = 3.2$$
$$h_C^0 = h_C + d_C^- + d_C^+ = 5 + T[3] + T[1] = 5 + 0.3 + 0.1 = 5.4$$
$$h_D^0 = h_D + d_D^- + d_D^+ = 2 + T[2] + T[2] = 2 + 0.2 + 0.2 = 2.4$$
$$h_E^0 = h_E + d_E^- + d_E^+ = 4 + T[1] + T[2] = 4 + 0.1 + 0.2 = 4.3$$

4

$$h_F^0 = h_F + d_F^- + d_F^+ = 6 + T[1] + T[1] = 6 + 0.1 + 0.1 = 6.2$$

b) Structural information helps the model distinguish similar graphs when considering only node-level features or simple neighbor aggregation, allowing the model to identify non-isomorphic graphs. Encoding structural properties like degree, subgraph structures, or motifs allows the model to identify such distinctions and increase the power of graph isomorphism tests as the Weisfeiler-Lehman (WL) test.

2024 가을 기말 10번 분자 학습에서 contrastive 의 한계, 왜 부분그래프 방법이 더 나은가

관련 (부분그래프)

문제

10. (10p) What are the limitations of contrastive learning methods in molecular structure learning? And why are the subgraph-based methods better at molecular structure learning?

3

공식 해설

10. (10p) What are the limitations of contrastive learning methods in molecular structure learning? And why are the subgraph-based methods better at molecular structure learning?

Contrastive methods rely heavily on data augmentations (e.g., node dropping, edge perturbation, subgraph sampling). However, random augmentations can destroy the molecular structures, and disrupt the chemical properties. Moreover, designing augmentations that preserve chemical validity is challenging for molecular graphs.

8

Subgraph-based methods explicitly extract and utilize localized structural patterns, making them better suited for molecular structure learning. Subgraph-based methods capture functional groups, rings, and other motifs, which are fundamental to molecular properties. That is, Subgraph-based methods can incorporate domain-specific knowledge about functional groups or reaction mechanisms, enhancing their relevance to molecular tasks. Moreover, Subgraph-based methods are inherently more interpretable because they explicitly focus on molecular substructures.

2025 가을 기말 3번 (2024 기말 10번과 같은 문제) 관련

문제

3. (Subgraph)(5p) What are the limitations of contrastive learning methods in molecular structure learning? And why are the subgraph-based methods better at molecular structure learning?

1

공식 해설

3. (Subgraph)(5p) Recently, various studies have used contrastive learning-based methods to learn molecular structures by employing graph augmentations (e.g., edge deletion) to generate multiple molecular views. What are the limitations of contrastive learning methods in molecular structure learning? And why are the subgraph-based methods better at molecular structure learning?

SOLUTIONS: Contrastive methods rely heavily on data augmentations (e.g., node dropping, edge perturbation, subgraph sampling). However, random augmentations can destroy the molecular structures and disrupt the chemical properties. Moreover, designing augmentations that preserve chemical validity is challenging for molecular graphs.

Subgraph-based methods explicitly extract and utilize localized structural patterns, making them better suited for molecular structure learning. Subgraph-based methods capture functional groups, rings, and other motifs, which are fundamental to molecular properties. That is, Subgraph-based methods can incorporate domain-specific knowledge about functional groups or reaction mechanisms, enhancing their relevance to molecular tasks. Moreover, Subgraph-based methods are inherently more interpretable because they explicitly focus on molecular substructures.

04 계산 그래프와 계산 부담, full batch

4주차, 금요일

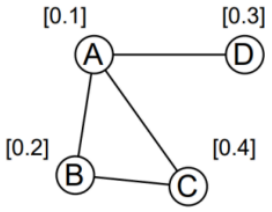
(a) 계산 그래프와 계산 부담: 노드마다 k-hop 계산 트리, 층이 깊어지면 지수적으로 커짐 (이웃 폭증), 허브 노드

(b) 왜 GCN 은 full batch 로만 도나: 이웃 수를 미리 알 수 없어 노드별 메모리를 못 잡음 → $AimesX$ 통째로, 차수 분포 (허브), 역전파 부하

2024 가을 기말 2번 GraphSAGE 계산 + How does GraphSAGE address the scalability problem? (b) 의 해법 쪽

문제

2. (10p) Consider an undirected graph G of four nodes A, B, C, and D given in the following figure. Each node has an initial feature that is the number standing next to it (i.e., the initial feature of node 'A' is $h_A^{(0)} = 0.1$). According to GraphSAGE model with a MAX aggregation function, the feature of a node i at layer k can be updated as:

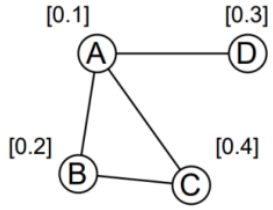


$$h_{N(i)}^{(k)} = \text{AGGREGATE}\left(\left\{h_u^{(k-1)}, \forall u \in N(i)\right\}\right)$$
$$h_i^{(k)} = \text{ReLU}\left(h_i^{(k-1)} || h_{N(i)}^{(k)}\right)$$

where $\|$ is a concatenation, $\text{ReLU}(x) = \max(0, x)$, $N(i)$ is the neighbour nodes of node i.

a) Calculate the feature of each node at $k = 1$.
b) How does the GraphSage model address the scalability problem?

2. (10p) Consider an undirected graph G of five nodes A, B, C, and D given in the following figure. Each node has initial features that are the numbers standing next to it (i.e., the initial feature of node 'A' is $h_A^{(0)} = 0.1$). According to GraphSAGE model with a MAX aggregation function, the feature of a node i at layer k can be updated as:



$$h_{N(i)}^{(k)} = \text{AGGREGATE}\left(\left\{h_u^{(k-1)}, \forall u \in N(i)\right\}\right)$$

$$h_i^{(k)} = \text{ReLU}\left(h_i^{(k-1)} \,\|\, h_{N(i)}^{(k)}\right)$$

where $\|$ is a concatenation, $\text{ReLU}(x) = \max(0, x)$, *N(i)* is the neighbour nodes of node *i*.

a) Calculate the feature of each node at $k = 1$.

b) How does the GraphSage model address the scalability problem?

SOLUTIONS:

a)

$$h_{N(i)}^{(k)} = \text{AGGREGATE}\left(\left\{h_u^{(k-1)}, \forall u \in N(i)\right\}\right)$$

$$h_{N(A)}^{(1)} = MAX\left(h_B, h_C, h_D\right) = MAX\left(0.4, 0.2, 0.3\right) = 0.4$$

$$h_{N(B)}^{(1)} = MAX\left(h_A, h_C\right) = MAX\left(0.1, 0.4\right) = 0.4$$

$$h_{N(C)}^{(1)} = MAX\left(h_A, h_B\right) = MAX\left(0.2, 0.1\right) = 0.2$$

$$h_{N(D)}^{(1)} = MAX\left(h_A\right) = MAX\left(0.1\right) = 0.1$$

$$h_i^{(k)} = \text{ReLU}\left(h_i^{(k-1)} \,\|\, h_{N(i)}^{(k)}\right)$$

$$h_A^{(1)} = \text{ReLU}\left(h_A^{(0)} \,\|\, h_{N(A)}^{(0)}\right) = Max\left(0, [0.1, 0.4]\right) = [0.1, 0.4]$$

$$h_B^{(1)} = \text{ReLU}\left(h_B^{(0)} \,\|\, h_{N(B)}^{(0)}\right) = Max\left(0, [0.2, 0.4]\right) = [0.2, 0.4]$$

$$h_C^{(1)} = \text{ReLU}\left(h_C^{(0)} \,\|\, h_{N(C)}^{(0)}\right) = Max\left(0, [0.4, 0.2]\right) = [0.4, 0.2]$$

$$h_D^{(1)} = \text{ReLU}\left(h_D^{(0)} \,\|\, h_{N(D)}^{(0)}\right) = Max\left(0, [0.3, 0.1]\right) = [0.3, 0.1]$$

b) GraphSAGE model addresses the scalability problem by employing a neighborhood sampling strategy and designing the model to operate inductively.

- GraphSAGE avoids processing the entire graph at once by sampling a fixed-size set of neighbors for each node. Instead of aggregating features from all neighbors (as done in GCNs), GraphSAGE samples a fixed number of *K* neighbors for each node.

- GraphSAGE is designed to learn inductive representations, meaning it can generate embeddings for unseen nodes or graphs without retraining.

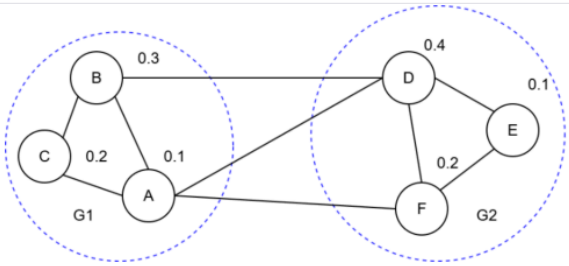
2025 가을 중간 5번 (a) What is mini-batch training?

(b) 와 이어짐

문제

5. (ClusterGCN) Consider an undirected graph G of six nodes A, B, C, D, E and F given in the following figure. The graph G contains two cluster G_1 and G_2. Each node has initial features that are the numbers standing next to it. According to ClusterGCN model, the feature of a node i at layer k can be updated as:

2

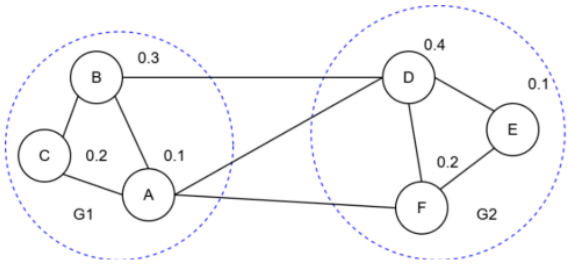


$$h_{N(i)}^{(k)} = \text{MEAN}\left(\left\{h_u^{(k-1)}, \forall u \in N(i), G_u = G_i\right\}\right)$$
$$h_i^{(k)} = \text{ReLU}\left(h_i^{(k-1)} || h_{N(i)}^{(k)}\right)$$
where || is a concatenation.

a. What is mini-batch training?
b. Calculate the output representations of all nodes at layer $k = 1$.

공식 해설

5. (ClusterGCN) (10p) Consider an undirected graph G of six nodes A, B, C, D, E and F given in the following figure. The graph G contains two cluster G_1 and G_2. Each node has initial features that are the numbers standing next to it. According to ClusterGCN model, the feature of a node i at layer k can be updated as:



$$h_{N(i)}^{(k)} = \text{MEAN}\left(\left\{h_u^{(k-1)}, \forall u \in N(i), G_u = G_i\right\}\right)$$
$$h_i^{(k)} = \text{ReLU}\left(h_i^{(k-1)} || h_{N(i)}^{(k)}\right)$$
where || is a concatenation.

a. (3p) What is mini-batch training?
   Mini-batch training means training on graph clusters (subgraphs) instead of the full graph, allowing efficient learning on large graphs while preserving local connectivity within each batch.
b. (7p) Calculate the output representations of all nodes at layer $k = 1$.

Ans:

4

$$h_{N(i)}^{(k)} = \text{MEAN}\left(\left\{h_u^{(k-1)}, \forall u \in N(i)\right\}, G_u = G_i\right)$$

$$h_{N(A)}^{(1)} = MEAN(h_B, h_C) = MEAN(0.3, 0.2) = 0.25$$

$$h_{N(B)}^{(1)} = MEAN(h_A, h_C) = MEAN(0.1,0.2) = 0.15$$

$$h_{N(C)}^{(1)} = MEAN(h_A, h_B) = MEAN(0.3,0.1) = 0.2$$

$$h_{N(D)}^{(1)} = MEAN(h_E, h_F) = MEAN(0.1,0.2) = 0.15$$

$$h_{N(E)}^{(1)} = MEAN(h_D, h_F) = MEAN(0.4,0.2) = 0.3$$

$$h_{N(F)}^{(1)} = MEAN(h_D, h_E) = MEAN(0.4,0.1) = 0.25$$

$$- - - -

05 Cluster-GCN 과 그 문제점

4주차, 금요일

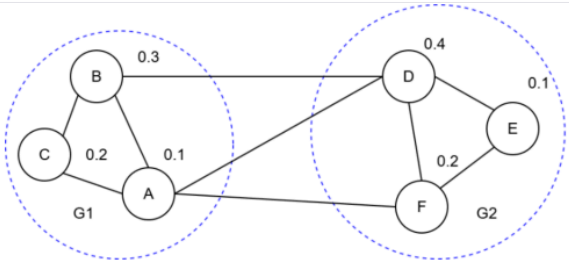
(a) Cluster-GCN 동작: 그래프를 클러스터로 나누고 (METIS), 클러스터 사이 엣지를 지우고, 클러스터 하나를 미니배치로. 손계산 가능
(b) 문제점: 클러스터 안 레이블이 비슷해 레이블 분포가 치우침 → 편향된 SGD, 지운 엣지의 정보 손실 (이질적 그래프, 넓게 봐야 하는 과제).
해법: 여러 클러스터를 무작위로 묶기

2025 가을 중간 5번 (a) 미니배치란 (b) ClusterGCN k=1 계산 **(a) 와 거의 그대로**

문제

5. (ClusterGCN) Consider an undirected graph G of six nodes A, B, C, D, E and F given in the following figure. The graph G contains two cluster G_1 and G_2. Each node has initial features that are the numbers standing next to it. According to ClusterGCN model, the feature of a node i at layer k can be updated as:

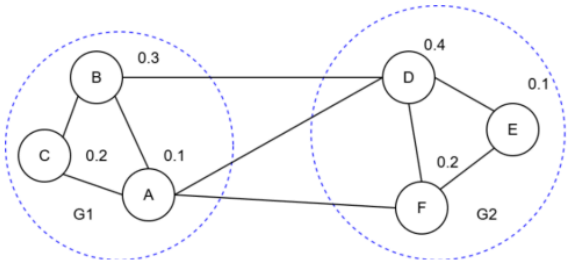
2



$$h_{N(i)}^{(k)} = \text{MEAN}\left(\left\{h_u^{(k-1)}, \forall u \in N(i), G_u = G_i\right\}\right)$$
$$h_i^{(k)} = \text{ReLU}\left(h_i^{(k-1)} || h_{N(i)}^{(k)}\right)$$
where || is a concatenation.

a. What is mini-batch training?
b. Calculate the output representations of all nodes at layer $k = 1$.

5. (ClusterGCN) (10p) Consider an undirected graph G of six nodes A, B, C, D, E and F given in the following figure. The graph G contains two cluster G_1 and G_2. Each node has initial features that are the numbers standing next to it. According to ClusterGCN model, the feature of a node i at layer k can be updated as:



$$h_{N(i)}^{(k)} = \text{MEAN}\left(\left\{h_u^{(k-1)}, \forall u \in N(i), G_u = G_i\right\}\right)$$
$$h_i^{(k)} = \text{ReLU}\left(h_i^{(k-1)} || h_{N(i)}^{(k)}\right)$$
where || is a concatenation.

a. (3p) What is mini-batch training?
Mini-batch training means training on graph clusters (subgraphs) instead of the full graph, allowing efficient learning on large graphs while preserving local connectivity within each batch.

b. (7p) Calculate the output representations of all nodes at layer $k = 1$.

Ans:

4

$$h_{N(i)}^{(k)} = \text{MEAN}\left(\left\{h_u^{(k-1)}, \forall u \in N(i)\right\}, G_u = G_i\right)$$
$$h_{N(A)}^{(1)} = MEAN(h_B, h_C) = MEAN(0.3, 0.2) = 0.25$$
$$h_{N(B)}^{(1)} = MEAN(h_A, h_C) = MEAN(0.1, 0.2) = 0.15$$
$$h_{N(C)}^{(1)} = MEAN(h_A, h_B) = MEAN(0.3, 0.1) = 0.2$$
$$h_{N(D)}^{(1)} = MEAN(h_E, h_F) = MEAN(0.1, 0.2) = 0.15$$
$$h_{N(E)}^{(1)} = MEAN(h_D, h_F) = MEAN(0.4, 0.2) = 0.3$$
$$h_{N(F)}^{(1)} = MEAN(h_D, h_E) = MEAN(0.4, 0.1) = 0.25$$
$$h_i^{(k)} = \text{ReLU}\left(h_i^{(k-1)} || h_{N(i)}^{(k)}\right)$$
$$h_A^{(1)} = \text{ReLU}\left(h_A^{(0)} || h_{N(A)}^{(1)}\right) = Max(0, [0.1, 0.25]) = [0.1, 0.25]$$
$$h_B^{(1)} = \text{ReLU}\left(h_B^{(0)} || h_{N(B)}^{(1)}\right) = Max(0, [0.3, 0.15]) = [0.3, 0.15]$$
$$h_C^{(1)} = \text{ReLU}\left(h_C^{(0)} || h_{N(C)}^{(1)}\right) = Max(0, [0.2, 0.25]) = [0.2, 0.2]$$
$$h_D^{(1)} = \text{ReLU}\left(h_D^{(0)} || h_{N(D)}^{(1)}\right) = Max(0, [0.4, 0.15]) = [0.4, 0.15]$$
$$h_E^{(1)} = \text{ReLU}\left(h_E^{(0)} || h_{N(E)}^{(1)}\right) = Max(0, [0.1, 0.3]) = [0.1, 0.3]$$
$$h_F^{(1)} = \text{ReLU}\left(h_F^{(0)} || h_{N(F)}^{(1)}\right) = Max(0, [0.2, 0.25]) = [0.2, 0.25]$$

고칠 점: 해설 h_C 줄 가운데 Max(0, [0.2, 0.25]) 는 [0.2, 0.2] 의 오타 (h_N(C) = 0.2). 마지막 답 [0.2, 0.2] 는 맞음

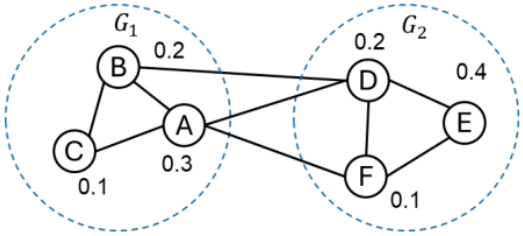
2024 가을 중간 3번 ClusterGCN k=1 계산

(a) 와 같은 꼴

문제

3. (10pt) Consider an undirected graph G of six nodes A, B, C, D, E and F given in the following figure. The graph G contains two cluster G_1 and G_2. Each node has initial features that are the numbers standing next to it. According to ClusterGCN model, the feature of a node i at layer k can be updated as:

1



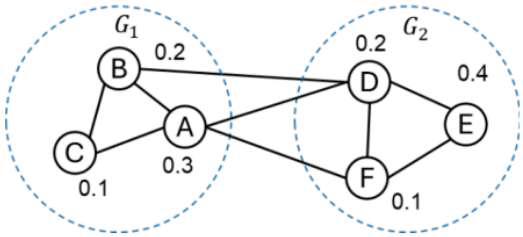
$$h_{N(i)}^{(k)} = \text{MEAN}\left(\left\{h_u^{(k-1)}, \forall u \in N(i), G_u = G_i\right\}\right)$$
$$h_i^{(k)} = \text{ReLU}\left(h_i^{(k-1)} || h_{N(i)}^{(k)}\right)$$

where || is a concatenation.

Calculate the output representations of all nodes at layer $k = 1$.

공식 해설

3. (10pt) Consider an undirected graph G of six nodes A, B, C, D, E and F given in the following figure. The graph G contains two cluster G_1 and G_2. Each node has initial features that are the numbers standing next to it. According to ClusterGCN model, the feature of a node i at layer k can be updated as:



$$h_{N(i)}^{(k)} = \text{MEAN}\left(\left\{h_u^{(k-1)}, \forall u \in N(i), G_u = G_i\right\}\right)$$
$$h_i^{(k)} = \text{ReLU}\left(h_i^{(k-1)} || h_{N(i)}^{(k)}\right)$$

where || is a concatenation.

Calculate the output representations of all nodes at layer $k = 1$.

Ans:

$$h_{N(i)}^{(k)} = \text{MEAN}\left(\left\{h_u^{(k-1)}, \forall u \in N(i)\right\}, G_u = G_i\right)$$
$$h_{N(A)}^{(1)} = MEAN(h_B, h_C) = MEAN(0.2, 0.1) = 0.15$$
$$h_{N(B)}^{(1)} = MEAN(h_A, h_C) = MEAN(0.3, 0.1) = 0.2$$

2

$h_{N(E)}^{(1)} = MEAN(h_D, h_F) = MEAN(0.2,0.1) = 0.15$

$h_{N(F)}^{(1)} = MEAN(h_D, h_E) = MEAN(0.4,0.2) = 0.3$

$h_i^{(k)} = \text{ReLU}\left(h_i^{(k-1)} || h_{N(i)}^{(k)}\right)$

$h_A^{(1)} = \text{ReLU}\left(h_A^{(0)} || h_{N(A)}^{(0)}\right) = Max(0, [0.3,0.15]) = [0.3,0.15]$

$h_B^{(1)} = \text{ReLU}\left(h_B^{(0)} || h_{N(B)}^{(0)}\right) = Max(0, [0.2,0.2]) = [0.2,0.2]$

$h_C^{(1)} = \text{ReLU}\left(h_C^{(0)} || h_{N(C)}^{(0)}\right) = Max(0, [0.1,0.25]) = [0.1,0.25]$

$h_D^{(1)} = \text{ReLU}\left(h_D^{(0)} || h_{N(D)}^{(0)}\right) = Max(0, [0.2,0.25]) = [0.2,0.25]$

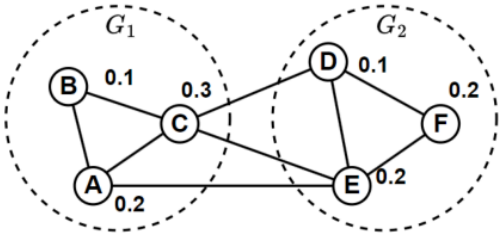
$h_E^{(1)} = \text{ReLU}\left(h_E^{(0)} || h_{N(E)}^{(0)}\right) = Max(0, [0.4,0.15]) = [0.4,0.15]$

$h_F^{(1)} = \text{ReLU}\left(h_F^{(0)} || h_{N(F)}^{(0)}\right) = Max(0, [0.1,0.25]) = [0.1,0.3]$

고칠 점: 해설 h_F 줄 가운데 [0.1, 0.25] 는 [0.1, 0.3] 의 오타 (답 [0.1, 0.3] 은 맞음). h_N(i) 의 위 첨자 (0) 은 (1) 이 맞음

2023 가을 기말 2번 ClusterGCN 으로 두 클러스터의 모든 노드 k=1 계산 (a) 와 같은 꼴 문제

2. Consider an undirected graph G of six nodes A, B, C, D, E and F given in the following figure. The graph G contains two cluster G_1 and G_2. Each node has initial features that are the numbers standing next to it. According to ClusterGCN model, the feature of a node i at layer k can be updated as:



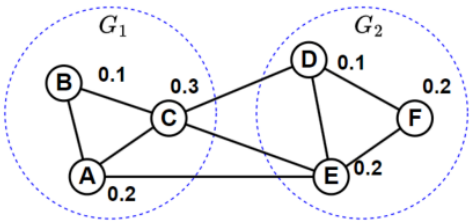
$$h_{N(i)}^{(k)} = \text{MEAN}\left(\left\{h_u^{(k-1)}, \forall u \in N(i), G_u = G_i\right\}\right)$$

$$h_i^{(k)} = \text{ReLU}\left(h_i^{(k-1)} \,\|\, h_{N(i)}^{(k)}\right)$$

where $\|$ is a concatenation.

Calculate the output representations of all nodes at layer $k = 1$.

2. Consider an undirected graph G of six nodes A, B, C, D, E and F given in the following figure. The graph G contains two cluster G_1 and G_2. Each node has initial features that are the numbers standing next to it. According to ClusterGCN model, the feature of a node i at layer k can be updated as:



$$h_{N(i)}^{(k)} = \text{MEAN}\left(\left\{h_u^{(k-1)}, \forall u \in N(i), G_u = G_i\right\}\right)$$

$$h_i^{(k)} = \text{ReLU}\left(h_i^{(k-1)} \,\|\, h_{N(i)}^{(k)}\right)$$

where $\|$ is a concatenation.

Calculate the output representations of all nodes at layer $k = 1$.

SOLUTIONS:

$$h_{N(i)}^{(k)} = \text{AGGREGATE}\left(\left\{h_u^{(k-1)}, \forall u \in N(i)\right\}, G_u = G_i\right)$$

$$h_{N(A)}^{(1)} = MEAN\left(h_B, h_C\right) = MEAN\left(0.1, 0.3\right) = 0.2$$

$$h_{N(B)}^{(1)} = MEAN\left(h_A, h_C\right) = MEAN\left(0.2, 0.3\right) = 0.25$$

$$h_{N(C)}^{(1)} = MEAN\left(h_A, h_B\right) = MEAN\left(0.2, 0.1\right) = 0.15$$

$$h_{N(D)}^{(1)} = MEAN\left(h_E, h_F\right) = MEAN\left(0.2, 0.2\right) = 0.2$$

$$h_{N(E)}^{(1)} = MEAN\left(h_D, h_F\right) = MEAN\left(0.1, 0.2\right) = 0.15$$

$$h_{N(F)}^{(1)} = MEAN\left(h_D, h_E\right) = MEAN\left(0.1, 0.2\right) = 0.15$$

$$h_i^{(k)} = \text{ReLU}\left(h_i^{(k-1)} \,\|\, h_{N(i)}^{(k)}\right)$$

$$h_A^{(1)} = \text{ReLU}\left(h_A^{(0)} \,\|\, h_{N(A)}^{(0)}\right) = Max\left(0, [0.2, 0.2]\right) = [0.2, 0.2]$$

$$h_B^{(1)} = \text{ReLU}\left(h_B^{(0)} \,\|\, h_{N(B)}^{(0)}\right) = Max\left(0, [0.1, 0.25]\right) = [0.1, 0.25]$$

$$h_C^{(1)} = \text{ReLU}\left(h_C^{(0)} \,\|\, h_{N(C)}^{(0)}\right) = Max\left(0, [0.3, 0.15]\right) = [0.3, 0.15]$$

$$h_D^{(1)} = \text{ReLU}\left(h_D^{(0)} \,\|\, h_{N(D)}^{(0)}\right) = Max\left(0, [0.1, 0.2]\right) = [0.1, 0.2]$$

$$h_E^{(1)} = \text{ReLU}\left(h_E^{(0)} \,\|\, h_{N(E)}^{(0)}\right) = Max\left(0, [0.2, 0.15]\right) = [0.2, 0.15]$$

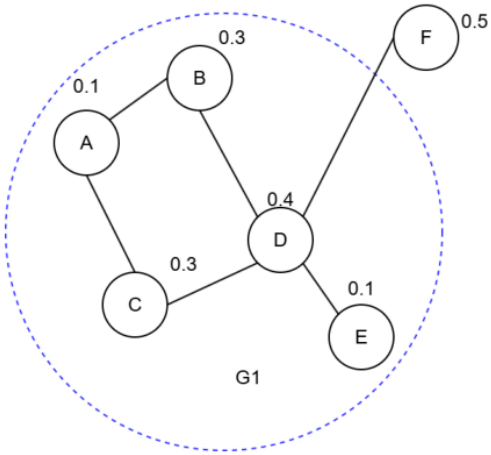
$$h_F^{(1)} = \text{ReLU}\left(h_E^{(0)} \,\|\, h_{N(F)}^{(0)}\right) = Max\left(0, [0.2, 0.15]\right) = [0.2, 0.15]$$

고칠 점: 해설 h_F 줄에 h_E(0) 이라고 적힌 곳은 h_F(0) 의 오타. h_N(i) 의 위 첨자도 (1) 이 맞음

2025 가을 중간 6번 GraphSAINT (Cluster-GCN 다음 모델) 계산 + 샘플링 차이 관련

문제

6. (GraphSAINT) Consider an undirected graph G of six nodes A, B, C, D, E and F given in the following figure. The graph G has subgraph sampling G_1. Each node has initial features that are the numbers standing next to it. According to GraphSAINT model, the feature of a node i at layer k can be updated as:

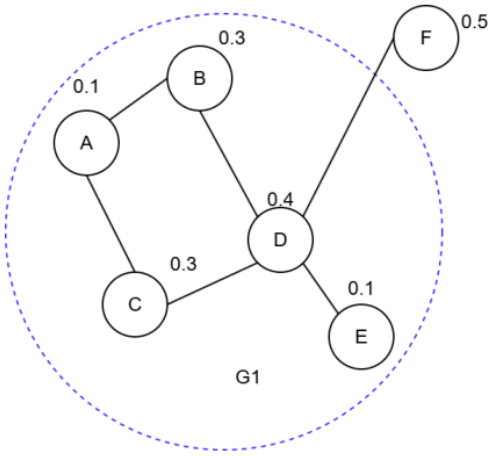


$$h_{N(i)}^{(k)} = \text{MEAN}\left(\left\{h_u^{(k-1)}, \forall u \in N(i), G_u = G_i\right\}\right)$$
$$h_i^{(k)} = \text{ReLU}\left(h_i^{(k-1)} || h_{N(i)}^{(k)}\right)$$
where || is a concatenation.

a. How does GraphSAINT's sampling strategy differ from node-wise or edge-wise sampling methods like GraphSAGE?
b. Calculate the output representations of nodes A, B, C, D, and E at layer $k = 1$.

3

6. (GraphSAINT) (10p) Consider an undirected graph G of six nodes A, B, C, D, E and F given in the following figure. The graph G has subgraph sampling G_1. Each node has initial features that are the numbers standing next to it. According to GraphSAINT model, the feature of a node i at layer k can be updated as:



$$h_{N(i)}^{(k)} = \text{MEAN}\left(\left\{h_u^{(k-1)}, \forall u \in N(i), G_u = G_i\right\}\right)$$
$$h_i^{(k)} = \text{ReLU}\left(h_i^{(k-1)} || h_{N(i)}^{(k)}\right)$$
where || is a concatenation.

a. (3p) How does GraphSAINT's sampling strategy differ from node-wise or edge-wise sampling methods like GraphSAGE?

5

GraphSAINT samples entire subgraphs for each mini-batch instead of individual nodes or edges, preserving more structural and neighborhood information while keeping training efficient

b. (7p) Calculate the output representations of nodes A, B, C, D, and E at layer $k = 1$.

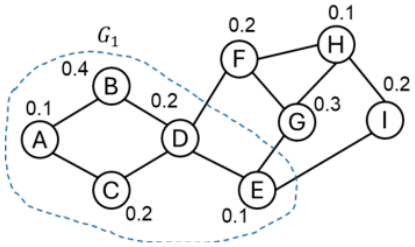
Ans:

$$h_{N(i)}^{(k)} = \text{MEAN}\left(\left\{h_u^{(k-1)}, \forall u \in N(i)\right\}, G_u = G_i\right)$$
$$h_{N(A)}^{(1)} = MEAN(h_B, h_C) = MEAN(0.3, 0.3) = 0.3$$
$$h_{N(B)}^{(1)} = MEAN(h_A, h_D) = MEAN(0.1, 0.4) = 0.25$$
$$h_{N(C)}^{(1)} = MEAN(h_A, h_D) = MEAN(0.1, 0.4) = 0.25$$
$$h_{N(D)}^{(1)} = MEAN(h_B, h_C, h_E) = MEAN(0.3, 0.3, 0.1) = 0.23$$
$$h_{N(E)}^{(1)} = MEAN(h_D) = MEAN(0.4) = 0.4$$
$$- -$$
$$h_i^{(k)} = \text{ReLU}\left(h_i^{(k-1)} || h_{N(i)}^{(k)}\right)$$
$$h_A^{(1)} = \text{ReLU}\left(h_A^{(0)} || h_{N(A)}^{(1)}\right) = Max(0, [0.1, 0.3]) = [0.1, 0.3]$$
$$h_B^{(1)} = \text{ReLU}\left(h_B^{(0)} || h_{N(B)}^{(1)}\right) = Max(0, [0.3, 0.25]) = [0.3, 0.25]$$
$$h_C^{(1)} = \text{ReLU}\left(h_C^{(0)} || h_{N(C)}^{(1)}\right) = Max(0, [0.3, 0.25]) = [0.3, 0.25]$$
$$h_D^{(1)} = \text{ReLU}\left(h_D^{(0)} || h_{N(D)}^{(1)}\right) = Max(0, [0.4, 0.23]) = [0.4, 0.23]$$
$$h_E^{(1)} = \text{ReLU}\left(h_E^{(0)} || h_{N(E)}^{(1)}\right) = Max(0, [0.1, 0.4]) = [0.1, 0.4]$$

2024 가을 중간 4번 GraphSAINT 계산 관련

문제

4. (10pt) Consider an undirected graph G of six nodes A, B, C, D, E and F given in the following figure. The graph G has subgraph sampling G_1. Each node has initial features that are the numbers standing next to it. According to GraphSAINT model, the feature of a node i at layer k can be updated as:



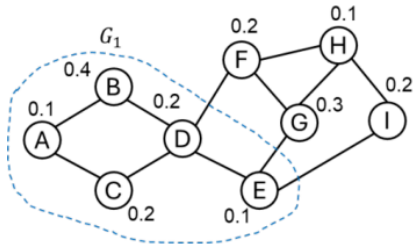
$$h_{N(i)}^{(k)} = \text{MEAN}\left(\left\{h_u^{(k-1)}, \forall u \in N(i), G_u = G_i\right\}\right)$$
$$h_i^{(k)} = \text{ReLU}\left(h_i^{(k-1)} || h_{N(i)}^{(k)}\right)$$

where || is a concatenation.

Calculate the output representations of all nodes at layer $k = 1$.

공식 해설

4. (10pt) Consider an undirected graph G of nine nodes A, B, C, D, E, F, G, H and I given in the following figure. The graph G has subgraph sampling G_1. Each node has initial features that are the numbers standing next to it. According to GraphSAINT model, the feature of a node i at layer k can be updated as:



$$h_{N(i)}^{(k)} = \text{MEAN}\left(\left\{h_u^{(k-1)}, \forall u \in N(i), G_u = G_i\right\}\right)$$
$$h_i^{(k)} = \text{ReLU}\left(h_i^{(k-1)} || h_{N(i)}^{(k)}\right)$$

where || is a concatenation.

Calculate the output representations of node A, B, C, D, E at layer $k = 1$.

Ans:

$$h_{N(i)}^{(k)} = \text{MEAN}\left(\left\{h_u^{(k-1)}, \forall u \in N(i)\right\}, G_u = G_i\right)$$
$$h_{N(A)}^{(1)} = MEAN(h_B, h_C) = MEAN(0.4, 0.2) = 0.3$$
$$h_{N(B)}^{(1)} = MEAN(h_A, h_D) = MEAN(0.1, 0.2) = 0.15$$
$$h_{N(C)}^{(1)} = MEAN(h_A, h_D) = MEAN(0.1, 0.2) = 0.15$$
$$h_{N(D)}^{(1)} = MEAN(h_B, h_C, h_E) = MEAN(0.4, 0.2, 0.1) = 0.23$$
$$h_{N(E)}^{(1)} = MEAN(h_D) = MEAN(0.2) = 0.2$$

3

$h_A^{(1)} = \text{ReLU}\left(h_A^{(0)} || h_{N(A)}^{(0)}\right) = Max(0, [0.1, 0.3]) = [0.1, 0.3]$

$$h_B^{(1)} = \text{ReLU}\left(h_B^{(0)} || h_{N(B)}^{(0)}\right) = Max(0, [0.15, 0.4]) = [0.15, 0.4]$$
$$h_C^{(1)} = \text{ReLU}\left(h_C^{(0)} || h_{N(C)}^{(0)}\right) = Max(0, [0.15, 0.2]) = [0.15, 0.2]$$
$$h_D^{(1)} = \text{ReLU}\left(h_D^{(0)} || h_{N(D)}^{(0)}\right) = Max(0, [0.23, 0.2]) = [0.23, 0.2]$$
$$h_E^{(1)} = \text{ReLU}\left(h_E^{(0)} || h_{N(E)}^{(0)}\right) = Max(0, [0.2, 0.1]) = [0.2, 0.1]$$

06 과평활, 과압축, under-reaching 과 저역통과 필터

5주차, 금요일

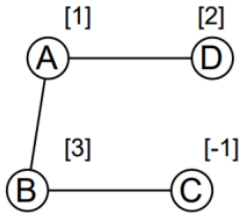
(a) 과평활, 과압축, under-reaching 의 정의와 드러나는 현상 (+ 수용 영역)

(b) 왜 GNN (이웃 평균) 은 저역통과 필터인가: 평균 = 블러, 이웃 사이 차이(고주파)를 층마다 깎음, 스펙트럼으로 $(1-\lambda_i)^l$

2024 가을 기말 3번 GCNII 계산 + (b) 과평활은 왜 생기나, GCNII 는 어떻게 푸나 **(a) 과평활 부분과 거의 그대로**

문제

3. (10p) Consider an undirected graph G of four nodes A, B, C, and D given in the following figure. Each node has an initial feature that is the number standing next to it, as follows:



The output of an GCNII model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma\left[\left((1-\beta)I_n\right)\cdot\left((1-\alpha)\tilde{A}\cdot H^{(k-1)} + \alpha H^{(0)}\right)\right]$$

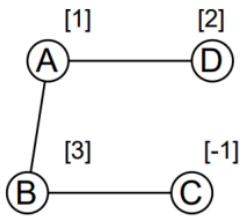
where $H^{(k)}$ denotes the output at layer k, $\tilde{A}$ is the normalized matrix $(\tilde{A} = D^{-1}A)$, I_n is the identity matrix, $\alpha = \beta = 0.5$, $\sigma(.)$ is a ReLU function $\mathrm{ReLU}(x) = \max(0, x)$.

a) Calculate the representations of each node at layer $k = 1$.

b) What are the reasons why the over-smoothing problem happens in graphs? And how does GCNII address the problem?

1

3. (10p) Consider an undirected graph G of five nodes A, B, C, and D given in the following figure. Each node has initial features that are the numbers standing next to it, as follows:



The output of an GCNII model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma\left[\left((1-\beta)I_n\right)\cdot\left((1-\alpha)\tilde{A}\cdot H^{(k-1)} + \alpha H^{(0)}\right)\right]$$

where $H^{(k)}$ denotes the output at layer k, $\tilde{A}$ is the normalized matrix ($\tilde{A} = D^{-1}A$), I_n is the identity matrix, $\alpha = \beta = 0.5$, σ is a ReLU function $\mathrm{ReLU}(x) = \max(0, x)$.

a) Calculate $\tilde{A}$ and the representations of each node at layer $k = 1$.
b) What are the reasons why the over-smoothing problem happens in graphs? And how does GCNII address the problem?

SOLUTIONS:

a)

$$\tilde{A} = D^{-1}A = \begin{bmatrix} 0 & 0.5 & 0 & 0.5 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$$H^{(k)} = \sigma\left[\left((1-\beta)I_n\right)\left((1-\alpha)\tilde{A}\cdot H^{(k-1)} + \alpha H^{(0)}\right)\right]$$

$$\tilde{A}\cdot H^{(0)} = \begin{bmatrix} 0 & 0.5 & 0 & 0.5 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}\cdot\begin{bmatrix} 1 \\ 3 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 0 \\ 3 \\ 1 \end{bmatrix}; I_n = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\left[\left((1-\alpha)\tilde{A}\cdot H^{(k-1)} + \alpha H^{(0)}\right)\cdot\left((1-\beta)I_n\right)\right]$$

$$= \left(0.5\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}\right)\cdot\left(0.5\begin{bmatrix} 2.5 \\ 0 \\ 3 \\ 1 \end{bmatrix} + 0.5\begin{bmatrix} 1 \\ 3 \\ -1 \\ 2 \end{bmatrix}\right) = 0.5\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}0.5\begin{bmatrix} 3.5 \\ 3 \\ 2 \\ 3 \end{bmatrix} = 0.25\begin{bmatrix} 3.5 \\ 3 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 0.875 \\ 0.75 \\ 0.5 \\ 0.75 \end{bmatrix}$$

$$H^{(k)} = \sigma\left(\begin{bmatrix} 0.875 \\ 0.75 \\ 0.5 \\ 0.75 \end{bmatrix}\right) = \begin{bmatrix} 0.875 \\ 0.75 \\ 0.5 \\ 0.75 \end{bmatrix}$$

b) Over-smoothing is a common problem in graph neural networks (GNNs), where node representations become indistinguishable as the number of layers increases. The reason is the excessive neighbor aggregation problem. That is, GNNs aggregate features from neighbors at each layer. As the number of layers increases, nodes receive information from increasingly distant nodes in the graph. As a result, this aggregation process leads to all nodes in a connected component having similar embeddings, causing a loss of discriminative power.

GCNII incorporates identity mapping by adding a weighted skip connection to the node embeddings:

- GCNII uses an initial residual connection that allows each layer to combine its output with the raw input features. This can helps maintain diversity in node representations by preserving raw node features across all layers. It also ensures that even as layers increase, the representations do not lose important local information.

- GCNII incorporates identity mapping to the weight matrix, preserving the initial node features across all layers, preventing the embeddings from collapsing.

고칠 점: 해설 첫 줄의 five nodes 는 네 노드의 오타. 답 [0.875, 0.75, 0.5, 0.75]^T 는 맞음

2025 가을 중간 7번 JKNet: (a) 과평활을 어떻게 완화하나 + 계산 관련 문제

7. (JK Network) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix} \qquad H^{(0)} = \begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 1 & 2 \\ -1 & 1 \\ 2 & 3 \end{bmatrix}$$

Assume that the output of an JK network model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \max\left(\sigma\left(\tilde{A} \cdot H^{(0)}\right), \sigma\left(\tilde{A} \cdot H^{(1)}\right), \ldots, \sigma\left(\tilde{A} \cdot H^{(k-1)}\right)\right)$$

where $H^{(k)}$ denotes the output at layer k, $\tilde{A}$ is the normalized matrix ($\tilde{A} = D^{-1}A$), σ is a ReLU function $\mathrm{ReLU}(x) = \max(0, x)$.

a) Explain how Jumping Knowledge Networks help mitigate the **over-smoothing problem** in deep GCNs.

b) Calculate $\tilde{A}$.

c) Calculate the output representations at layer $k = 2$.

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7. (JK Network) (10p) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix} \qquad H^{(0)} = \begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 1 & 2 \\ -1 & 1 \\ 2 & 3 \end{bmatrix}$$

Assume that the output of an JK network model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \max\left(\sigma\left(\tilde{A} \cdot H^{(0)}\right), \sigma\left(\tilde{A} \cdot H^{(1)}\right), \ldots, \sigma\left(\tilde{A} \cdot H^{(k-1)}\right)\right)$$

where $H^{(k)}$ denotes the output at layer k, $\tilde{A}$ is the normalized matrix ($\tilde{A} = D^{-1}A$), σ is a ReLU function $\mathrm{ReLU}(x) = \max(0, x)$.

a) (2p) Explain how Jumping Knowledge Networks help mitigate the **over-smoothing problem** in deep GCNs.

JK networks mitigate over-smoothing by combining representations from multiple layers, allowing each node to adaptively use information from both shallow and deep layers, preserving feature diversity

b) (4p) Calculate $\tilde{A}$.

c) (4p) Calculate the output representations at layer $k = 2$.

Ans:

$$D = \begin{bmatrix} 3 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}, D^{-1} = \begin{bmatrix} 1/3 & 0 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 0 & 1/2 \end{bmatrix}$$

$$\tilde{A} = D^{-1}A = \begin{bmatrix} 1/3 & 1/2 & 0 & 1/3 & 0 \\ 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 \\ 0 & 1/2 & 0 & 1/2 & 0 \end{bmatrix}$$

$$H^{(k)} = \max\left(\sigma(\tilde{A} \cdot H^{(0)}), \sigma(\tilde{A} \cdot H^{(1)}), \dots, \sigma(\tilde{A} \cdot H^{(k)})\right)$$

- Layer 1:

$$H^1 = \sigma(AH^0) = \text{ReLU}\left(\begin{bmatrix} 1/3 & 1/2 & 0 & 1/3 & 0 \\ 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 \\ 0 & 1/2 & 0 & 1/2 & 0 \end{bmatrix} \times \begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 1 & 2 \\ -1 & 1 \\ 2 & 3 \end{bmatrix}\right)$$

$$= \text{ReLU}\left(\begin{bmatrix} -1/6 & -1/6 \\ 3/2 & 5/2 \\ 0 & 0 \\ 1/2 & 2 \\ 0 & 0 \end{bmatrix}\right) = \begin{bmatrix} 0 & 0 \\ 3/2 & 5/2 \\ 0 & 0 \\ 1/2 & 2 \\ 0 & 0 \end{bmatrix}$$

- Layer 2:

$$H^{(2)} = \max\left(\sigma\left(\tilde{A}\cdot H^{(0)}\right), \sigma\left(\tilde{A}\cdot H^{(1)}\right)\right)$$

$$= \max\left(\sigma\left(\begin{bmatrix} \frac{1}{3} & \frac{1}{2} & 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{bmatrix}\begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 1 & 2 \\ -1 & 1 \\ 2 & 3 \end{bmatrix}\right), \sigma\left(\begin{bmatrix} \frac{1}{3} & \frac{1}{2} & 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 \\ \frac{3}{2} & \frac{5}{2} \\ 0 & 0 \\ \frac{1}{2} & 2 \\ 0 & 0 \end{bmatrix}\right)\right)$$

$$= \max\left(\sigma\begin{bmatrix} -1/6 & -1/6 \\ 3/2 & 5/2 \\ 0 & 0 \\ 1/2 & 2 \\ 0 & 0 \end{bmatrix}, \sigma\begin{bmatrix} \frac{11}{12} & \frac{23}{12} \\ 0 & 0 \\ 1 & \frac{9}{4} \\ \frac{1}{4} & 1 \\ 1 & \frac{9}{4} \end{bmatrix}\right) = \begin{bmatrix} \frac{11}{12} & \frac{23}{12} \\ 3/2 & 5/2 \\ 1 & \frac{9}{4} \\ \frac{1}{2} & 2 \\ 1 & \frac{9}{4} \end{bmatrix}$$

2025 가을 중간 9번 DeepGCN: (a) 깊어질 때 과평활을 어떻게 완화하나 + 계산 관련 문제

9. (DeepGCN) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \quad H^{(0)} = \begin{bmatrix} 0 & -1 \\ -1 & 3 \\ 2 & -1 \\ 0 & -3 \end{bmatrix}$$

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Assume that the hidden layer of an DeepGCNs model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma\left(A \cdot H^{(k-1)}\right) + H^{(k-1)},$$

where $H^{(k)}$ denotes the output at layer k, σ is a ReLU function $\mathrm{ReLU}(x) = \max(0, x)$.

a. How does DeepGCN mitigate the **over-smoothing problem** as depth increases?
b. Calculate the output of the GCN model at layer $k = 2$.

9. (DeepGCN) (10p) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \quad H^{(0)} = \begin{bmatrix} 0 & -2 \\ -1 & 3 \\ 4 & -2 \\ 0 & -5 \end{bmatrix}$$

Assume that the hidden layer of an DeepGCNs model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma\left(A \cdot H^{(k-1)}\right) + H^{(k-1)},$$

where $H^{(k)}$ denotes the output at layer k, σ is a ReLU function $\mathrm{ReLU}(x) = \max(0, x)$.

a. (3p) How does DeepGCN mitigate the **over-smoothing problem** as depth increases?
DeepGCN mitigates over-smoothing by using residual connections, normalization layers, and dilated/dynamic aggregation, which help preserve feature diversity and stable information flow in deep layers

b. (7p) Calculate the output of the GCN model at layer $k = 2$.
Ans:

$$H^{(1)} = \sigma\left(AH^{(0)}\right) + H^{(0)} = \mathrm{ReLU}\left(\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & -2 \\ -1 & 3 \\ 4 & -2 \\ 0 & -5 \end{bmatrix}\right) + \begin{bmatrix} 0 & -2 \\ -1 & 3 \\ 4 & -2 \\ 0 & -5 \end{bmatrix}$$

$$= \mathrm{ReLU}\left(\begin{bmatrix} 7 & -7 \\ -1 & 3 \\ 3 & -7 \\ 3 & 1 \end{bmatrix}\right) + \begin{bmatrix} 0 & -2 \\ -1 & 3 \\ 4 & -2 \\ 0 & -5 \end{bmatrix} = \begin{bmatrix} 7 & 0 \\ 0 & 3 \\ 3 & 0 \\ 3 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -2 \\ -1 & 3 \\ 4 & -2 \\ 0 & -5 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & -2 \\ -2 & 6 \\ 7 & -2 \\ 3 & -4 \end{bmatrix}$$

$$H^{(2)} = \sigma\left(AH^{(1)}\right) + H^{(1)} = \mathrm{ReLU}\left(\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 7 & -2 \\ -2 & 6 \\ 7 & -2 \\ 3 & -4 \end{bmatrix}\right) + \begin{bmatrix} 7 & -2 \\ -2 & 6 \\ 7 & -2 \\ 3 & -4 \end{bmatrix}$$

$$= \mathrm{ReLU}\left(\begin{bmatrix} 10 & -6 \\ -2 & 6 \\ 10 & -6 \\ 5 & 4 \end{bmatrix}\right) + \begin{bmatrix} 7 & -2 \\ -2 & 6 \\ 7 & -2 \\ 3 & -4 \end{bmatrix} = \begin{bmatrix} 10 & 0 \\ 0 & 6 \\ 10 & 0 \\ 5 & 4 \end{bmatrix} + \begin{bmatrix} 7 & -2 \\ -2 & 6 \\ 7 & -2 \\ 3 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} 17 & -2 \\ -2 & 12 \\ 17 & -2 \\ 8 & 0 \end{bmatrix}$$

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고칠 점: 해설지의 H(0) 가 문제지와 달라요. 해설지 H(0) 로도 AH(0) 1행은 [4, -7] (해설 [7, -7]), 3행은 [0, -7] (해설 [3, -7]), H(1) 2행은 [-1, 6] (해설 [-2, 6]). 맞는 H(2) 는 해설지 숫자로 [[11, -2], [-1, 12], [11, -2], [6, 0]], 문제지 숫자로 [[5, -1], [-1, 12], [5, -1], [2, 4]]

07 GCNII: initial residual, identity mapping

5주차, 토요일

(a) GCNII 두 장치와 식 (손계산 가능): $\sigma\left[((1-\beta)I_n)((1-\alpha)\tilde{A}H+\alpha H^{(0)})\right]$

(b) 그 장치들이 왜 과평활을 막고 더 깊게 쌓게 하나

2025 가을 중간 8번 (a) GCNII 두 장치 (b) Ã (c) H(1) 계산 **거의 그대로**

문제

8. (GCNII) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A=\begin{bmatrix}0&1&1&0&0\\1&1&0&1&1\\1&0&1&0&0\\1&1&1&1&1\\0&1&0&0&1\end{bmatrix} \qquad H^{(0)}=\begin{bmatrix}3\\-1\\1\\2\\3\end{bmatrix}$$

Assume that the output of an GCNII model of all nodes at layer (k) can be calculated as:

$$H^{(k)}=\sigma\left[\left((1-\beta)I_n\right)\cdot\left((1-\alpha)\tilde{A}\cdot H^{(k-1)}+\alpha H^{(0)}\right)\right]$$

where $H^{(k)}$ denotes the output at layer k, $\tilde{A}$ is the normalized matrix ($\tilde{A}=D^{-1}A$), I_n is the identity matrix, $\alpha=\beta=0.5$, σ is a ReLU function $\mathrm{ReLU}(x)=\max(0,x)$.

a) What are the two major components introduced in GCNII to overcome the depth limitation problem in GCNs?

b) Calculate $\tilde{A}$.

c) Calculate the output representations at layer $k=1$.

공식 해설

8. (GCNII) (10p) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A=\begin{bmatrix}0&1&1&0&0\\1&1&0&1&1\\1&0&1&0&0\\1&1&1&1&1\\0&1&0&0&1\end{bmatrix} \qquad H^{(0)}=\begin{bmatrix}3\\-1\\1\\2\\3\end{bmatrix}$$

Assume that the output of an GCNII model of all nodes at layer (k) can be calculated as:

$$H^{(k)}=\sigma\left[\left((1-\beta)I_n\right)\cdot\left((1-\alpha)\tilde{A}\cdot H^{(k-1)}+\alpha H^{(0)}\right)\right]$$

where $H^{(k)}$ denotes the output at layer k, $\tilde{A}$ is the normalized matrix ($\tilde{A}=D^{-1}A$), I_n is the identity matrix, $\alpha=\beta=0.5$, σ is a ReLU function $\mathrm{ReLU}(x)=\max(0,x)$.

a) (2p) What are the two major components introduced in GCNII to overcome the depth limitation problem in GCNs?

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GCNII introduces initial residual connections and identity mapping to preserve original features and enable deeper, more stable GCN training

b) (4p) Calculate $\tilde{A}$.

c) (4p) Calculate the output representations at layer $k = 1$

Ans:

$$D = \begin{bmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}, D^{-1} = \begin{bmatrix} 1/2 & 0 & 0 & 0 & 0 \\ 0 & 1/4 & 0 & 0 & 0 \\ 0 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1/5 & 0 \\ 0 & 0 & 0 & 0 & 1/2 \end{bmatrix}$$

$$\tilde{A} = D^{-1}A = \begin{bmatrix} 1/2 & 0 & 0 & 0 & 0 \\ 0 & 1/4 & 0 & 0 & 0 \\ 0 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1/5 & 0 \\ 0 & 0 & 0 & 0 & 1/2 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0.5 & 0.5 & 0 & 0 \\ 0.25 & 0.25 & 0 & 0.25 & 0.25 \\ 0.5 & 0 & 0.5 & 0 & 0 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0 & 0.5 & 0 & 0 & 0.5 \end{bmatrix}$$

- $H^{(k)} = \sigma\left[\left((1-\beta)I_n\right)\left((1-\alpha)\tilde{A} \cdot H^{(k-1)} + \alpha H^{(0)}\right)\right]$

$$\tilde{A} \cdot H^{(k-1)} = \begin{bmatrix} 0 & 0.5 & 0.5 & 0 & 0 \\ 0.25 & 0.25 & 0 & 0.25 & 0.25 \\ 0.5 & 0 & 0.5 & 0 & 0 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0 & 0.5 & 0 & 0 & 0.5 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ -1 \\ 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1.75 \\ 2 \\ 1.6 \\ 1 \end{bmatrix}$$

$$I_n = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\left[\left((1-\beta)I_n\right)\left((1-\alpha)\tilde{A} \cdot H^{(k-1)} + \alpha H^{(0)}\right)\right]$$

$$= \left(0.5 \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}\right) \cdot \left(0.5 \begin{bmatrix} 0 \\ 1.75 \\ 2 \\ 1.6 \\ 1 \end{bmatrix} + 0.5 \begin{bmatrix} 3 \\ -1 \\ 1 \\ 2 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 1.5 \\ 0.375 \\ 1.5 \\ 1.8 \\ 2 \end{bmatrix}$$

$$H^{(k)} = \sigma\left(\begin{bmatrix} 1.5 \\ 0.375 \\ 1.5 \\ 1.8 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 1.5 \\ 0.375 \\ 1.5 \\ 1.8 \\ 2 \end{bmatrix}$$

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고칠 점: 해설지가 마지막 (1-β) = 0.5 곱하기를 빠뜨려 답이 두 배. 맞는 답은 [0.75, 0.1875, 0.75, 0.9, 1]^T

2024 가을 중간 6번 Ã, H(1) 계산 **(a) 와 거의 그대로**

문제

6. (15pt) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \end{bmatrix} \qquad H^{(0)} = \begin{bmatrix} 3 \\ -2 \\ 2 \\ 4 \\ 3 \end{bmatrix}$$

Assume that the output of an GCNII model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma\left[\left((1-\beta)I_n\right) \cdot \left((1-\alpha)\tilde{A} \cdot H^{(k-1)} + \alpha H^{(0)}\right)\right]$$

where $H^{(k)}$ denotes the output at layer k, $\tilde{A}$ is the normalized matrix ($\tilde{A} = D^{-1}A$), I_n is the identity matrix, $\alpha = \beta = 0.5$, σ is a ReLU function $\mathrm{ReLU}(x) = \max(0, x)$.

a) Calculate $\tilde{A}$.

b) Calculate the output representations at layer $k = 1$.

공식 해설

6. (15pt) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \end{bmatrix} \qquad H^{(0)} = \begin{bmatrix} 3 \\ -2 \\ 2 \\ 4 \\ 3 \end{bmatrix}$$

Assume that the output of an GCNII model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma\left[\left((1-\beta)I_n\right) \cdot \left((1-\alpha)\tilde{A} \cdot H^{(k-1)} + \alpha H^{(0)}\right)\right]$$

where $H^{(k)}$ denotes the output at layer k, $\tilde{A}$ is the normalized matrix ($\tilde{A} = D^{-1}A$), I_n is the identity matrix, $\alpha = \beta = 0.5$, σ is a ReLU function $\mathrm{ReLU}(x) = \max(0, x)$.

c) Calculate $\tilde{A}$.

d) Calculate the output representations at layer $k = 1$.

Ans:

$$D = \begin{bmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{bmatrix}$$

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$$D^{-1} = \begin{bmatrix} 0 & \frac{1}{5} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{5} & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{3} \end{bmatrix}$$

$$\tilde{A} = D^{-1}A = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{5} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{5} & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} \\ 1 & 0 & 0 & 0 & 0 \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} \\ 0 & \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \end{bmatrix}$$

- $H^{(k)} = \sigma\left[\left((1-\beta)I_n\right)\left((1-\alpha)\tilde{A} \cdot H^{(k-1)} + \alpha H^{(0)}\right)\right]$

$$\tilde{A} \cdot H^{(k-1)} = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} \\ 1 & 0 & 0 & 0 & 0 \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} \\ 0 & \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \end{bmatrix} \cdot \begin{bmatrix} 3 \\ -2 \\ 2 \\ 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 3 \\ 2 \\ 1 \end{bmatrix}$$

$$I_n = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\left[\left((1-\beta)I_n\right)\left((1-\alpha)\tilde{A} \cdot H^{(k-1)} + \alpha H^{(0)}\right)\right]$$

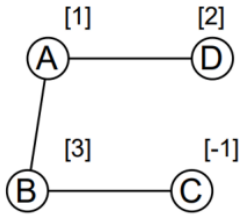
$$= \left(0.5 \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}\right) \cdot \left(0.5 \begin{bmatrix} 0 \\ 2 \\ 3 \\ 2 \\ 1 \end{bmatrix} + 0.5 \begin{bmatrix} 3 \\ -2 \\ 2 \\ 4 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} \frac{3}{4} \\ 0 \\ \frac{5}{4} \\ \frac{3}{2} \\ 1 \end{bmatrix}$$

$$H^{(k)} = \sigma\left(\begin{bmatrix} 0 \\ \frac{5}{4} \\ \frac{3}{2} \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ \frac{5}{4} \\ \frac{3}{2} \\ 1 \end{bmatrix}$$

2024 가을 기말 3번 GCNII 계산 + 과평활 이유와 GCNII 의 해법 **(a)(b) 둘 다 거의 그대로**

문제

3. (10p) Consider an undirected graph G of four nodes A, B, C, and D given in the following figure. Each node has an initial feature that is the number standing next to it, as follows:



The output of an GCNII model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma\left[\left((1-\beta)I_n\right)\cdot\left((1-\alpha)\tilde{A}\cdot H^{(k-1)} + \alpha H^{(0)}\right)\right]$$

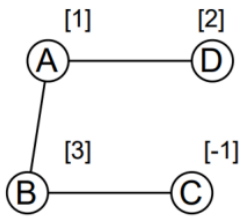
where $H^{(k)}$ denotes the output at layer k, $\tilde{A}$ is the normalized matrix ($\tilde{A} = D^{-1}A$), I_n is the identity matrix, $\alpha = \beta = 0.5$, $\sigma(.)$ is a ReLU function $\mathrm{ReLU}(x) = \max(0, x)$.

a) Calculate the representations of each node at layer $k = 1$.

b) What are the reasons why the over-smoothing problem happens in graphs? And how does GCNII address the problem?

1

3. (10p) Consider an undirected graph G of five nodes A, B, C, and D given in the following figure. Each node has initial features that are the numbers standing next to it, as follows:



The output of an GCNII model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma\left[\left((1-\beta)I_n\right)\cdot\left((1-\alpha)\tilde{A}\cdot H^{(k-1)} + \alpha H^{(0)}\right)\right]$$

where $H^{(k)}$ denotes the output at layer k, $\tilde{A}$ is the normalized matrix ($\tilde{A} = D^{-1}A$), I_n is the identity matrix, $\alpha = \beta = 0.5$, σ is a ReLU function $\mathrm{ReLU}(x) = \max(0, x)$.

a) Calculate $\tilde{A}$ and the representations of each node at layer $k = 1$.
b) What are the reasons why the over-smoothing problem happens in graphs? And how does GCNII address the problem?

SOLUTIONS:

a)

$$\tilde{A} = D^{-1}A = \begin{bmatrix} 0 & 0.5 & 0 & 0.5 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$$H^{(k)} = \sigma\left[\left((1-\beta)I_n\right)\left((1-\alpha)\tilde{A}\cdot H^{(k-1)} + \alpha H^{(0)}\right)\right]$$

$$\tilde{A}\cdot H^{(0)} = \begin{bmatrix} 0 & 0.5 & 0 & 0.5 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}\cdot\begin{bmatrix} 1 \\ 3 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 0 \\ 3 \\ 1 \end{bmatrix}; I_n = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\left[\left((1-\alpha)\tilde{A}\cdot H^{(k-1)} + \alpha H^{(0)}\right)\cdot\left((1-\beta)I_n\right)\right]$$

$$= \left(0.5\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}\right)\cdot\left(0.5\begin{bmatrix} 2.5 \\ 0 \\ 3 \\ 1 \end{bmatrix} + 0.5\begin{bmatrix} 1 \\ 3 \\ -1 \\ 2 \end{bmatrix}\right) = 0.5\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}0.5\begin{bmatrix} 3.5 \\ 3 \\ 2 \\ 3 \end{bmatrix} = 0.25\begin{bmatrix} 3.5 \\ 3 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 0.875 \\ 0.75 \\ 0.5 \\ 0.75 \end{bmatrix}$$

$$H^{(k)} = \sigma\left(\begin{bmatrix} 0.875 \\ 0.75 \\ 0.5 \\ 0.75 \end{bmatrix}\right) = \begin{bmatrix} 0.875 \\ 0.75 \\ 0.5 \\ 0.75 \end{bmatrix}$$

b) Over-smoothing is a common problem in graph neural networks (GNNs), where node representations become indistinguishable as the number of layers increases. The reason is the excessive neighbor aggregation problem. That is, GNNs aggregate features from neighbors at each layer. As the number of layers increases, nodes receive information from increasingly distant nodes in the graph. As a result, this aggregation process leads to all nodes in a connected component having similar embeddings, causing a loss of discriminative power.

GCNII incorporates identity mapping by adding a weighted skip connection to the node embeddings:

- GCNII uses an initial residual connection that allows each layer to combine its output with the raw input features. This can helps maintain diversity in node representations by preserving raw node features across all layers. It also ensures that even as layers increase, the representations do not lose important local information.

- GCNII incorporates identity mapping to the weight matrix, preserving the initial node features across all layers, preventing the embeddings from collapsing.

고칠 점: 해설 첫 줄의 five nodes 는 네 노드의 오타. 답 [0.875, 0.75, 0.5, 0.75]^T 는 맞음. (b) 해설이 identity mapping 을 'preserving the initial node features' 라고 적어 initial residual 과 섞였어요. initial residual = 처음 특징 H(0) 를 섞기, identity mapping = 가중치 W 를 단위행렬 I 쪽으로 당기기

2024 가을 중간 7번 DeepGCN (residual) k=2 계산 관련 (다른 스킵 연결과 비교)

문제

7. (10pt) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix} \qquad H^{(0)} = \begin{bmatrix} 0 & -2 \\ -1 & 3 \\ 4 & -2 \\ 0 & -5 \end{bmatrix}$$

Assume that the hidden layer of an DeepGCNs model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma\left(A \cdot H^{(k-1)}\right) + H^{(k-1)},$$

where $H^{(k)}$ denotes the output at layer k, σ is a ReLU function $\mathrm{ReLU}(x) = \max(0, x)$. Calculate the output of the GCN model at layer $k = 2$.

7. (10pt) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix} \quad H^{(0)} = \begin{bmatrix} 0 & -2 \\ -1 & 3 \\ 4 & -2 \\ 0 & -5 \end{bmatrix}$$

Assume that the hidden layer of an DeepGCNs model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma\left(A \cdot H^{(k-1)}\right) + H^{(k-1)},$$

where $H^{(k)}$ denotes the output at layer k, σ is a ReLU function $\mathrm{ReLU}(x) = \max(0, x)$. Calculate the output of the GCN model at layer $k = 2$.

Ans:

$$H^{(1)} = \sigma\left(AH^{(0)}\right) + H^{(0)} = \mathrm{ReLU}\left(\begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}\begin{bmatrix} 0 & -2 \\ -1 & 3 \\ 4 & -2 \\ 0 & -5 \end{bmatrix}\right) + \begin{bmatrix} 0 & -2 \\ -1 & 3 \\ 4 & -2 \\ 0 & -5 \end{bmatrix}$$

$$= \max\left(\begin{bmatrix} -1 & -2 \\ -1 & 3 \\ -1 & -4 \\ 3 & -4 \end{bmatrix}\right) + \begin{bmatrix} 0 & -2 \\ -1 & 3 \\ 4 & -2 \\ 0 & -5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 3 \\ 0 & 0 \\ 3 & 0 \end{bmatrix} + \begin{bmatrix} 0 & -2 \\ -1 & 3 \\ 4 & -2 \\ 0 & -5 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -2 \\ -1 & 6 \\ 4 & -2 \\ 3 & -5 \end{bmatrix}$$

$$H^{(2)} = \sigma\left(AH^{(1)}\right) + H^{(1)} = \mathrm{ReLU}\left(\begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}\begin{bmatrix} 0 & -2 \\ -1 & 6 \\ 4 & -2 \\ 3 & -5 \end{bmatrix}\right) + \begin{bmatrix} 0 & -2 \\ -1 & 6 \\ 4 & -2 \\ 3 & -5 \end{bmatrix}$$

$$= \max\left(\begin{bmatrix} 2 & 1 \\ -1 & 6 \\ 2 & -1 \\ 6 & -1 \end{bmatrix}\right) + \begin{bmatrix} 0 & -2 \\ -1 & 6 \\ 4 & -2 \\ 3 & -5 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 0 & 6 \\ 2 & 0 \\ 6 & 0 \end{bmatrix} + \begin{bmatrix} 0 & -2 \\ -1 & 6 \\ 4 & -2 \\ 3 & -5 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 12 \\ 6 & -2 \\ 9 & -5 \end{bmatrix}$$

08 동질성 지표 (homophily metrics)

6주차, 토요일

(a) 지표 정의와 계산: edge, node, class, adjusted homophily (+ CPM, LI, structural)
(b) 지표마다 무엇을 재고 어떤 특징/의미 차이가 있나 (전체 엣지 vs 노드별 평균 vs 클래스 불균형 보정 vs 차수 보정)

기출 없음. 6주차 (이질성) 는 2026 에 새로 들어온 내용이에요. 사이트 카드에 직접 만든 손계산 예제가 있어요.

09

H2GCN

6주차, 토요일

(a) H2GCN 동작: $H^{(0)} = \mathrm{ReLU}(XW_0)$, 자기 루프 없이 1홉 S_1, 2홉 S_2 따로 집계, 모든 층 결과 이어 붙이기 (손계산 가능)
(b) 세 설계 (D1 자기/이웃 분리, D2 고차 이웃, D3 중간 표현 결합) 가 왜 이질성을 개선하나

기출 없음. 6주차 (이질성) 는 2026 에 새로 들어온 내용이에요. 사이트 카드에 직접 만든 손계산 예제가 있어요.

10 GAT 어텐션과 GAT 대 GATv2

7주차, 일요일

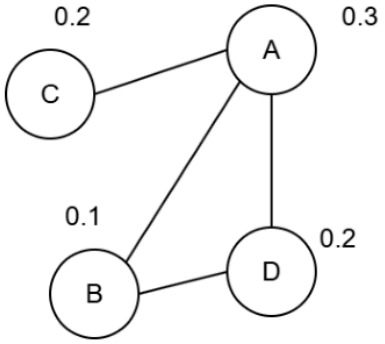
(a) GAT 어텐션 동작 (손계산): 점수 e, 정규화 α, 가중합

(b) GAT 대 GATv2: 순서 차이 (정적 vs 동적 어텐션) 와 그 이유

2025 가을 중간 10번 (a) GAT 가 푸는 GCN 의 문제 (b) e_AB, e_AC, e_AD (c) h_A **(a) 와 거의 그대로**

문제

10. (GAT) Consider an undirected graph G of four nodes A, B, C, and D given in the following figure. Each node has initial features that are the numbers standing next to it (i.e., the initial feature of node 'A' is $h_A^{(0)} = 0.3$). According to GAT model, the weight matrix W is randomly initialized as [0.5]. The feature of node 'i' at layer (k) can be updated as:



$$h_i^{(k)} = \sigma\left(\sum_{m \in N(i)} \alpha_{im} W h_m\right)$$

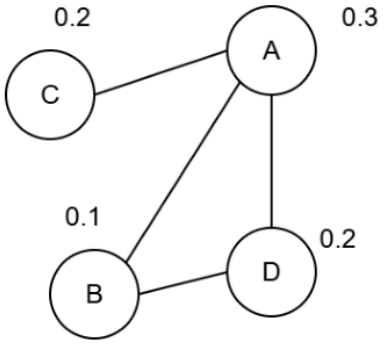
Where $\alpha_{im} = \frac{e_{im}}{\sum_{k \in N(i)} e_{ik}}$ and

$e_{im} = \sigma\big(\text{MEAN}(Wh_i, Wh_m)\big)$

σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

a) What problem in traditional GCNs does GAT aim to solve?
b) Calculate the attention coefficients e_{AB}, e_{AC}, and e_{AD}
c) Calculate the feature of node 'A' at $k = 1$.

10. (GAT) (10p) Consider an undirected graph G of four nodes A, B, C, and D given in the following figure. Each node has initial features that are the numbers standing next to it (i.e., the initial feature of node 'A' is $h_A^{(0)} = 0.3$). According to GAT model, the weight matrix W is randomly initialized as [0.5]. The feature of node 'i' at layer (k) can be updated as:



$$h_i^{(k)} = \sigma\left(\sum_{m \in N(i)} \alpha_{im} W h_m\right)$$

Where $\alpha_{im} = \frac{e_{im}}{\sum_{k \in N(i)} e_{ik}}$ and

$e_{im} = \sigma\big(\text{MEAN}(Wh_i, Wh_m)\big)$

σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

a) (2p) What problem in traditional GCNs does GAT aim to solve?
GAT solves the problem of equal weighting in traditional GCNs, it lets the model learn different importance weights for each neighbor instead of treating all neighbors equally

b) (4p) Calculate the attention coefficients e_{AB}, e_{AC}, and e_{AD}

c) (4p) Calculate the feature of node 'A' at $k = 1$.

Ans:

$$h_i^{(k)} = \sigma\left(\sum_{m \in N(i)} \alpha_{im} W h_m\right)$$

where: $\alpha_{im} = \frac{e_{im}}{\sum_{k \in N(i)} e_{ik}}$, and $e_{im} = \sigma\big(\text{MEAN}(Wh_i, Wh_m)\big)$

$$e_{AC} = \sigma\big(\text{MEAN}(Wh_A, Wh_C)\big) = \sigma\big(\text{MEAN}(0.5 * 0.3, 0.5 * 0.2)\big)$$
$$= \sigma\big(\text{MEAN}(0.15, 0.1)\big) = 0.125$$
$$e_{AB} = \sigma\big(\text{MEAN}(Wh_A, Wh_B)\big) = \sigma\big(\text{MEAN}(0.5 * 0.3, 0.5 * 0.1)\big)$$
$$= \sigma\big(\text{MEAN}(0.15, 0.05)\big) = 0.1$$
$$e_{AD} = \sigma\big(\text{MEAN}(Wh_A, Wh_D)\big) = \sigma\big(\text{MEAN}(0.5 * 0.3, 0.5 * 0.2)\big)$$
$$= \sigma\big(\text{MEAN}(0.15, 0.1)\big) = 0.125$$

$$\alpha_{AB} = \frac{e_{AB}}{e_{AB} + e_{AC} + e_{AD}} = \frac{0.1}{0.125 + 0.1 + 0.125} = 0.286$$
$$\alpha_{AC} = \alpha_{AD} = 0.357$$

$$h_i^{(k)} = \sigma\left(\sum_{m \in N(i)} \alpha_{im} W h_m\right) = \sigma(\alpha_{AB} W h_B + \alpha_{AC} W h_C + \alpha_{AD} W h_D)$$
$$= \sigma(0.286 * 0.5 * 0.1 + 0.357 * 0.5 * 0.2 + 0.357 * 0.5 * 0.2) = 0.0857$$

2025 가을 중간 11번 (a) GATv2 가 고친 GAT 의 한계 (b) 두 e_ij 비교 계산 **(b) 와 거의 그대로**

문제

11. (GATv2)
    a. What is the key limitation of GAT does GATv2 aim to fix?
    b. Calculate attention score e_{ij} to compare different between GAT and GATv2, given

    GAT $e_{ij} = LeakyReLU(a(W\overrightarrow{h_i}, W\overrightarrow{h_j}))$

    GATv2 $e_{ij} = aLeakyReLU(W[\overrightarrow{h_i}, \overrightarrow{h_j}])$

    $$\overrightarrow{h_i} = \begin{bmatrix}1\\2\end{bmatrix}, \overrightarrow{h_j} = \begin{bmatrix}1\\-1\end{bmatrix}, W = \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}, a = [1,1,1,1]$$

    LeakyReLU activation, where $\begin{cases} x = x, if\ x > 0 \\ x = 0.02x, if\ x < 0 \end{cases}$

공식 해설

11. (GATv2) (5p)
    a. (2p) What is the key limitation of GAT does GATv2 aim to fix?
    GATv2 fixes GAT's limitation of limited expressiveness by making the attention mechanism dynamic and nonlinear, allowing it to better capture complex relationships between node pairs.

    b. (3p) Calculate attention score e_{ij} to compare different between GAT and GATv2, given

    GAT $e_{ij} = LeakyReLU(a(W\overrightarrow{h_i}, W\overrightarrow{h_j}))$

    GATv2 $e_{ij} = aLeakyReLU(W[\overrightarrow{h_i}, \overrightarrow{h_j}])$

    $$\overrightarrow{h_i} = \begin{bmatrix}1\\2\end{bmatrix}, \overrightarrow{h_j} = \begin{bmatrix}1\\-1\end{bmatrix}, W = \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}, a = [1,1,1,1]$$

    LeakyReLU activation, where $\begin{cases} x = x, if\ x > 0 \\ x = 0.02x, if\ x < 0 \end{cases}$

    Ans:

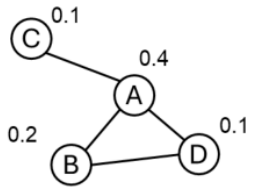
    GAT $e_{ij} = LeakyReLU\left(a(W\overrightarrow{h_i}, W\overrightarrow{h_j})\right) = LeakyReLU([1,1,1,1][1,2,1,-1] = 3$

    GATv2 $e_{ij} = aLeakyReLU(W[\overrightarrow{h_i}, \overrightarrow{h_j}]) = [1,1,1,1]LeakyReLU([1,2,1,-1]) = [1,1,1,1][1,2,1,-0.02] = 3.98$

2024 가을 중간 8번 GAT e, h_A 계산 (a) 와 같은 꼴

문제

8. (15pt) Consider an undirected graph G of four nodes A, B, C, and D given in the following figure. Each node has initial features that are the numbers standing next to it (i.e., the initial feature of node 'A' is $h_A^{(0)} = 0.4$). According to GAT model, the weight matrix W is randomly initialized as [0.5]. The feature of node 'i' at layer (k) can be updated as:



$$h_i^{(k)} = \sigma\left(\sum_{m \in N(i)} \alpha_{im} W h_m\right)$$

Where $\alpha_{im} = \frac{e_{im}}{\sum_{k \in N(i)} e_{ik}}$ and

$e_{im} = \sigma(\text{MEAN}(Wh_i, Wh_m))$

σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

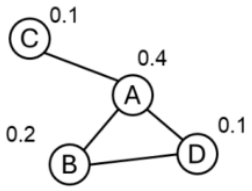
a) Calculate the attention coefficients e_{AB}, e_{AC}, and e_{AD}

3

b) Calculate the feature of node ‘A’ at $k = 1$.

공식 해설

8. (15pt) Consider an undirected graph G of four nodes A, B, C, and D given in the following figure. Each node has initial features that are the numbers standing next to it (i.e., the initial



$$h_i^{(k)} = \sigma\left(\sum_{m \in N(i)} \alpha_{im} W h_m\right)$$

Where $\alpha_{im} = \frac{e_{im}}{\sum_{k \in N(i)} e_{ik}}$ and

$e_{im} = \sigma\big(\text{MEAN}(Wh_i, Wh_m)\big)$

σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

a) Calculate the attention coefficients e_{AB}, e_{AC}, and e_{AD}

b) Calculate the feature of node ‘A’ at $k = 1$.

Ans:

$$h_i^{(k)} = \sigma\left(\sum_{m \in N(i)} \alpha_{im} W h_m\right)$$

where: $\alpha_{im} = \frac{e_{im}}{\sum_{k \in N(i)} e_{ik}}$, and $e_{im} = \sigma\big(\text{MEAN}(Wh_i, Wh_m)\big)$

$e_{AC} = \sigma\big(\text{MEAN}(Wh_A, Wh_C)\big) = \sigma\big(\text{MEAN}(0.5 * 0.4, 0.5 * 0.1)\big)$

$= \sigma\big(\text{MEAN}(0.2, 0.05)\big) = 0.125$

$e_{AB} = \sigma\big(\text{MEAN}(Wh_A, Wh_B)\big) = \sigma\big(\text{MEAN}(0.5 * 0.4, 0.5 * 0.2)\big)$

$= \sigma\big(\text{MEAN}(0.2, 0.1)\big) = 0.15$

$e_{AD} = \sigma\big(\text{MEAN}(Wh_A, Wh_D)\big) = \sigma\big(\text{MEAN}(0.5 * 0.4, 0.5 * 0.1)\big)$

$= \sigma\big(\text{MEAN}(0.2, 0.05)\big) = 0.125$

$$\alpha_{AB} = \frac{e_{AB}}{e_{AB} + e_{AC} + e_{AD}} = \frac{0.15}{0.125 + 0.15 + 0.125} = 0.375$$

$$\alpha_{AC} = \alpha_{AD} = 0.3125$$

$$h_i^{(k)} = \sigma\left(\sum_{m \in N(i)} \alpha_{im} W h_m\right) = \sigma(\alpha_{AB} W h_B + \alpha_{AC} W h_C + \alpha_{AD} W h_D)$$

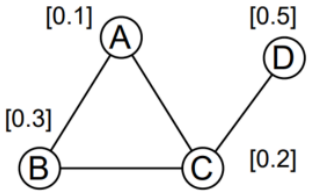
$$= \sigma(0.375 * 0.5 * 0.2 + 0.3125 * 0.5 * 0.1 + 0.3125 * 0.5 * 0.1) = 0.06875$$

2024 가을 기말 4번 GAT 계산 (W = 0.1) + 어텐션은 노드 중요도 문제를 어떻게 푸나

(a) 와 같은 꼴

문제

4. (10p) Consider an undirected graph G of four nodes A, B, C, and D given in the following figure. Each node has an initial feature that is the number standing next to it. According to GAT model, the weight matrix W is initialized as [0.1]. The feature of node 'i' at layer (k) can be updated as:



$$h_i^{(k)} = \sigma\left(\sum_{m\in N(i)} \alpha_{im} W h_m\right)$$

$$\text{where: } \alpha_{im} = \frac{e_{im}}{\sum_{k\in N(i)} e_{ik}}$$

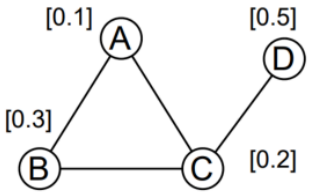
$$\text{and } e_{im} = \sigma\big(\text{MEAN}(Wh_i, Wh_m)\big)$$

σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

a) Calculate the feature of node 'A' at $k = 1$.
b) How can the attention mechanism address the node importance problem?

공식 해설

4. (10p) Consider an undirected graph G of four nodes A, B, C, and D given in the following figure. Each node has initial features that are the numbers standing next to it. According to GAT model, the weight matrix W is initialized as [0.1]. The feature of node 'i' at layer (k) can be updated as:



$$h_i^{(k)} = \sigma\left(\sum_{m\in N(i)} \alpha_{im} W h_m\right)$$

$$\text{where: } \alpha_{im} = \frac{e_{im}}{\sum_{k\in N(i)} e_{ik}}, \text{ and } e_{im} = \sigma\big(\text{MEAN}\big(Wh_i, Wh_m\big)\big)$$

σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

a) Calculate the attention coefficients e_{AB} and e_{AC}. Then, calculate the feature of node 'A' at $k = 1$.
b) How can the attention mechanism address the node importance problem?

SOLUTIONS:

4

$$h_i^{(k)} = \sigma\left(\sum_{m \in N(i)} \alpha_{im} W h_m\right)$$

where: $\alpha_{im} = \dfrac{e_{im}}{\sum_{k \in N(i)} e_{ik}}$, and $e_{im} = \sigma\left(\text{MEAN}\left(Wh_i, Wh_m\right)\right)$

$$e_{AC} = \sigma\left(\text{MEAN}\left(Wh_A, Wh_C\right)\right) = \sigma\left(\text{MEAN}\left(0.1*0.1, 0.1*0.2\right)\right)$$

$$= \sigma\left(\text{MEAN}\left(0.01, 0.02\right)\right) = 0.015$$

$$e_{AB} = \sigma\left(\text{MEAN}\left(Wh_A, Wh_B\right)\right) = \sigma\left(\text{MEAN}\left(0.1*0.1, 0.1*0.3\right)\right)$$

$$= \sigma\left(\text{MEAN}\left(0.01, 0.03\right)\right) = 0.02$$

$$\alpha_{AB} = \frac{e_{AB}}{e_{AB} + e_{AC}} = \frac{0.02}{0.02 + 0.015} = 0.57$$

$$\alpha_{AC} = 0.43$$

$$h_A^{(1)} = \sigma\left(\sum_{m \in N(i)} \alpha_{Am} W h_m\right) = \sigma\left(\alpha_{AB} W h_B + \alpha_{AC} W h_C\right)$$

$$= \sigma\left(0.57*0.1*0.3 + 0.43*0.1*0.2\right)$$

$$= \text{Re}LU\left(0.57*0.1*0.3 + 0.43*0.1*0.2\right)$$

$$= 0.0171 + 0.0086 = 0.0257$$

b) The attention mechanism assigns a unique weight (attention score) to each neighboring node based on its relevance to the target node. The relevance is computed using a learned function that evaluates the relationship between the target node's features and each neighbor's features. That is, neighbors that are more relevant to target node will receive higher weights , and their features contribute more during aggregation. Meanwhile, less important or irrelevant neighbors contribute less, reducing noise in the aggregated representation.

11 HAN: 동작과 어텐션을 두 번 하는 이유

7주차, 일요일

(a) HAN 동작 (손계산): 메타패스 나열, 노드 수준 어텐션 계수, P_1 특징

(b) 왜 어텐션을 두 번 (노드 수준 + 의미 수준) 하나

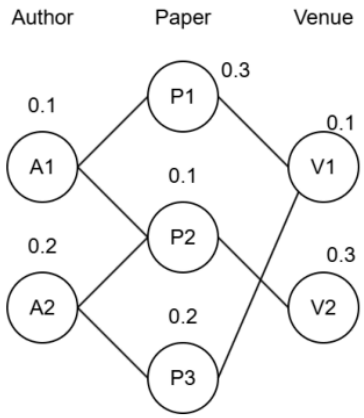
2025 가을 중간 12번 (a) 이종 그래프란 (b) 메타패스 PAP, PVP 나열 + 계수 (c) P1 특징 **(a) 와 거의 그대로**

문제

12. (HAN) Consider a heterogeneous graph given in the following figure. There are three types of nodes in the academic network: Author (A), Paper (P), and Venue (V). Each node has initial features that are the numbers standing next to it (i.e., the initial feature of node 'A_1' is

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$h_{A_1}^{(0)} = 0.2$). According to HAN model, the weight matrix W is randomly initialized as [0.5]. The feature of node 'i' at layer (k) can be updated as:



$$h_i^{(k)} = \sigma\left(\sum_{m \in N(i)} \alpha_{im}^{\Phi} W h_m\right)$$

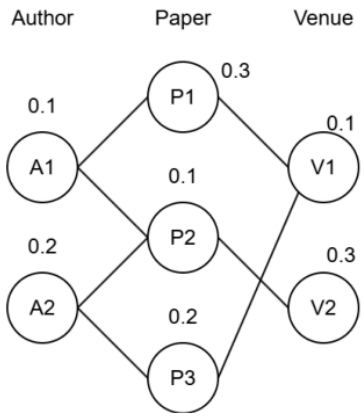
Where $\alpha_{im}^{\Phi} = \frac{e_{im}^{\Phi}}{\sum_{k \in N^{\Phi}(i)} e_{ik}^{\Phi}}$ and

$$e_{im}^{\Phi} = \sigma\left(\text{MEAN}\left(W h_i^{\Phi}, W h_m^{\Phi}\right)\right)$$

σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

a) What is heterogeneous graph?
b) List all the meta-path PAP and PVP. Calculate the attention coefficients of each meta-path PAP and PVP.
c) Calculate the feature of node 'P_1' at $k = 1$.

12. (HAN) (5p) Consider a heterogeneous graph given in the following figure. There are three types of nodes in the academic network: Author (A), Paper (P), and Venue (V). Each node has initial features that are the numbers standing next to it (i.e., the initial feature of node 'A_1' is $h_{A_1}^{(0)} = 0.2$). According to HAN model, the weight matrix W is randomly initialized as [0.5]. The feature of node 'i' at layer (k) can be updated as:



$$h_i^{(k)} = \sigma\left(\sum_{m \in N(i)} \alpha_{im}^{\Phi} W h_m\right)$$

Where $\alpha_{im}^{\Phi} = \frac{e_{im}^{\Phi}}{\sum_{k \in N^{\Phi}(i)} e_{ik}^{\Phi}}$ and

$$e_{im}^{\Phi} = \sigma\left(\mathrm{MEAN}\left(W h_i^{\Phi}, W h_m^{\Phi}\right)\right)$$

σ is a ReLU function $\mathrm{ReLU}(x) = \max(0, x)$.

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a) (1p) What is heterogeneous graph?
Heterogeneous graph is a graph that has different types of node.

b) (2p) List all the meta-path PAP and PVP. Calculate the attention coefficients of each meta-path PAP and PVP.

c) (2p) Calculate the feature of node 'P_1' at $k = 1$.

Ans:

List meta-path:

- PAP: $P_1A_1P_2, P_2A_2P_3$.
- PVP: $P_1V_1P_3$.

$$h_i^{(k)} = \sigma\left(\sum_{m\in N(i)} \alpha_{im}^{\Phi} W h_m\right)$$

Where $\alpha_{im}^{\Phi} = \frac{e_{im}^{\Phi}}{\sum_{k\in N^{\Phi}(i)} e_{ik}^{\Phi}}$ and $e_{im}^{\Phi} = \sigma\left(\text{MEAN}\left(Wh_i^{\Phi}, Wh_m^{\Phi}\right)\right)$

PAP:

$$e_{P_1P_2}^{\Phi} = \sigma\left(\text{MEAN}\left(Wh_{P_1}, Wh_{P_2}\right)\right) = \sigma\left(\text{MEAN}(0.5 * 0.3, 0.5 * 0.1)\right)$$
$$= \sigma\left(\text{MEAN}(0.15, 0.05)\right) = 0.1$$
$$e_{P_2P_3}^{\Phi} = \sigma\left(\text{MEAN}\left(Wh_{P_2}, Wh_{P_3}\right)\right) = \sigma\left(\text{MEAN}(0.5 * 0.1, 0.5 * 0.2)\right)$$
$$= \sigma\left(\text{MEAN}(0.05, 0.1)\right) = 0.075$$

$$e_{P_1P_3}^{\Phi} = \sigma\left(\text{MEAN}\left(Wh_{P_1}, Wh_{P_3}\right)\right) = \sigma\left(\text{MEAN}(0.5 * 0.3, 0.5 * 0.2)\right)$$
$$= \sigma\left(\text{MEAN}(0.15, 0.1)\right) = 0.125$$

$$e_{P_1A_1}^{\Phi} = \sigma\left(\text{MEAN}\left(Wh_{P_1}, Wh_{A_1}\right)\right) = \sigma\left(\text{MEAN}(0.5 * 0.3, 0.5 * 0.1)\right)$$
$$= \sigma\left(\text{MEAN}(0.15, 0.05)\right) = 0.1$$

$$e_{A_1P_2}^{\Phi} = \sigma\left(\text{MEAN}\left(Wh_{A_1}, Wh_{P_2}\right)\right) = \sigma\left(\text{MEAN}(0.5 * 0.1, 0.5 * 0.1)\right)$$
$$= \sigma\left(\text{MEAN}(0.05, 0.05)\right) = 0.05$$

$$\alpha_{P_1P_2}^{\Phi} = \frac{e_{P_1P_2}^{\Phi}}{e_{P_1P_2}^{\Phi} + e_{P_1P_3}^{\Phi}} = \frac{0.1}{0.1 + 0.125} \approx 0.444$$
$$\alpha_{P_1P_3}^{\Phi} = \frac{e_{P_1P_3}^{\Phi}}{e_{P_1P_2}^{\Phi} + e_{P_1P_3}^{\Phi}} = \frac{0.125}{0.1 + 0.125} \approx 0.555$$
$$\alpha_{P_2P_3}^{\Phi} = 0.075$$

$$\alpha^{\Phi}_{P_1A_1} = \frac{e^{\Phi}_{P_1A_1}}{e^{\Phi}_{P_1A_1}} = 1 = \alpha^{\Phi}_{A_1P_2} = \alpha^{\Phi}_{P_1V_1} = \alpha^{\Phi}_{V_1P_3}$$

PAP: $P_1A_1P_2$

$$h_i^{(k)} = \sigma\left(\sum_{m\in N(i)} \alpha^{\Phi}_{im} W h_m\right) = \sigma\left(\left(\alpha^{\Phi}_{P_1A_1} W h_{A_1}\right) + \left(\alpha^{\Phi}_{A_1P_2} W h_{P_2}\right)\right)$$

$$= \sigma\big((1 * 0.5 * 0.1) + (1 * 0.5 * 0.1)\big) = 0.1$$

PVP: $P_1V_1P_3$

$$h_i^{(k)} = \sigma\left(\sum_{m\in N(i)} \alpha^{\Phi}_{im} W h_m\right) = \sigma\left(\left(\alpha^{\Phi}_{P_1V_1} W h_{V_1}\right) + \left(\alpha^{\Phi}_{V_1P_3} W h_{P_3}\right)\right) =$$

$$= \sigma\big((1 * 0.5 * 0.1) + (1 * 0.5 * 0.2)\big) = 0.15$$

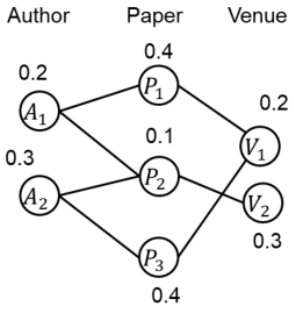
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고칠 점: 해설이 조금 엉성하다: α_P2P3 칸에 나누기 전 e 값 (0.075) 을 적었고 (나누면 0.429), α 를 P1 의 두 메타패스 이웃을 합쳐 나눴고, P1 특징은 메타패스 경로를 따라 더하는 방식. 문제 글의 h_A1 = 0.2 는 그림 (0.1) 과 달라요, 해설은 그림 값으로 풂

2024 가을 중간 9번 메타패스 나열 + 계수 + P1 특징 (숫자만 다름) **(a) 와 거의 그대로**

문제

9. (10pt) Consider a heterogeneous graph given in the following figure. There are three types of nodes in the academic network: Author (A), Paper (P), and Venue (V). Each node has initial features that are the numbers standing next to it (i.e., the initial feature of node 'A_1' is $h_{A_1}^{(0)} = 0.2$). According to HAN model, the weight matrix W is randomly initialized as [0.5]. The feature of node 'i' at layer (k) can be updated as:



$$h_i^{(k)} = \sigma\left(\sum_{m \in N(i)} \alpha_{im}^{\Phi} W h_m\right)$$

Where $\alpha_{im}^{\Phi} = \frac{e_{im}^{\Phi}}{\sum_{k \in N^{\Phi}(i)} e_{ik}^{\Phi}}$ and

$$e_{im}^{\Phi} = \sigma\left(\mathrm{MEAN}\left(W h_i^{\Phi}, W h_m^{\Phi}\right)\right)$$

σ is a ReLU function $\mathrm{ReLU}(x) = \max(0, x)$.

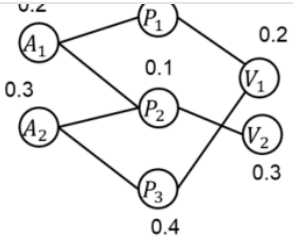
a) List all the meta-path PAP and PVP. Calculate the attention coefficients of each meta-path PAP and PVP.
b) Calculate the feature of node 'P_1' at $k = 1$.

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공식 해설

9. (10pt) Consider a heterogeneous graph given in the following figure. There are three types of nodes in the academic network: Author (A), Paper (P), and Venue (V). Each node has initial features that are the numbers standing next to it (i.e., the initial feature of node 'A_1' is $h_{A_1}^{(0)} = 0.2$). According to HAN model, the weight matrix W is randomly initialized as [0.5]. The feature of node 'i' at layer (k) can be updated as:

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$$\left(\overline{m\in N(i)}\right)$$

Where $\alpha_{im}^{\Phi} = \frac{e_{im}^{\Phi}}{\sum_{k\in N^{\Phi}(i)} e_{ik}^{\Phi}}$ and

$$e_{im}^{\Phi} = \sigma\left(\text{MEAN}\left(Wh_i^{\Phi}, Wh_m^{\Phi}\right)\right)$$

σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

a) List all the meta-path PAP and PVP. Calculate the attention coefficients $\alpha_{P_1P_2}^{\Phi}, \alpha_{P_1P_3}^{\Phi}, and\ \alpha_{P_2P_3}^{\Phi}$.
b) Calculate the feature of node 'P_1' at $k = 1$.

Ans:

List meta-path:
- PAP: $P_1A_1P_2, P_2A_2P_3$.
- PVP: $P_1V_1P_3$.

$$h_i^{(k)} = \sigma\left(\sum_{m\in N(i)} \alpha_{im}^{\Phi} Wh_m\right)$$

Where $\alpha_{im}^{\Phi} = \frac{e_{im}^{\Phi}}{\sum_{k\in N^{\Phi}(i)} e_{ik}^{\Phi}}$ and $e_{im}^{\Phi} = \sigma\left(\text{MEAN}\left(Wh_i^{\Phi}, Wh_m^{\Phi}\right)\right)$

PAP:

$$e_{P_1P_2}^{\Phi} = \sigma\left(\text{MEAN}\left(Wh_{P_1}, Wh_{P_2}\right)\right) = \sigma\left(\text{MEAN}(0.5 * 0.4, 0.5 * 0.1)\right)$$
$$= \sigma\left(\text{MEAN}(0.2, 0.05)\right) = 0.125$$
$$e_{P_2P_3}^{\Phi} = \sigma\left(\text{MEAN}\left(Wh_{P_2}, Wh_{P_3}\right)\right) = \sigma\left(\text{MEAN}(0.5 * 0.1, 0.5 * 0.4)\right)$$
$$= \sigma\left(\text{MEAN}(0.05, 0.2)\right) = 0.125$$

$$e_{P_1P_3}^{\Phi} = \sigma\left(\text{MEAN}\left(Wh_{P_1}, Wh_{P_3}\right)\right) = \sigma\left(\text{MEAN}(0.5 * 0.4, 0.5 * 0.4)\right)$$
$$= \sigma\left(\text{MEAN}(0.2, 0.2)\right) = 0.2$$

$$e_{P_1A_1}^{\Phi} = \sigma\left(\text{MEAN}\left(Wh_{P_1}, Wh_{A_1}\right)\right) = \sigma\left(\text{MEAN}(0.5 * 0.4, 0.5 * 0.2)\right)$$
$$= \sigma\left(\text{MEAN}(0.2, 0.1)\right) = 0.15$$

$$e_{A_1P_2}^{\Phi} = \sigma\left(\text{MEAN}\left(Wh_{A_1}, Wh_{P_2}\right)\right) = \sigma\left(\text{MEAN}(0.5 * 0.2, 0.5 * 0.1)\right)$$
$$= \sigma\left(\text{MEAN}(0.1, 0.05)\right) = 0.075$$

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$$\alpha^{\Phi}_{P_1P_3} = \frac{8}{13} \approx 0.6154$$

$$\alpha^{\Phi}_{P_2P_3} = 0.125$$

$$\alpha^{\Phi}_{P_1A_1} = \frac{e^{\Phi}_{P_1A_1}}{e^{\Phi}_{P_1A_1}} = 1 = \alpha^{\Phi}_{A_1P_2} = \alpha^{\Phi}_{P_1V_1} = \alpha^{\Phi}_{V_1P_3}$$

PAP: $P_1A_1P_2$

$$\alpha^{\Phi}_{P_1A_1} = \frac{e^{\Phi}_{P_1A_1}}{e^{\Phi}_{P_1P_2} + e^{\Phi}_{P_1P_3}}$$

$$h_i^{(k)} = \sigma\left(\sum_{m\in N(i)} \alpha^{\Phi}_{im} W h_m\right) = \sigma\left(\left(\alpha^{\Phi}_{P_1A_1} W h_{A_1}\right) + \left(\alpha^{\Phi}_{A_1P_2} W h_{P_2}\right)\right)$$

$$= \sigma\big((1 * 0.5 * 0.2) + (1 * 0.5 * 0.1)\big) = 0.015$$

PVP: $P_1V_1P_3$

$$h_i^{(k)} = \sigma\left(\sum_{m\in N(i)} \alpha^{\Phi}_{im} W h_m\right) = \sigma\left(\left(\alpha^{\Phi}_{P_1V_1} W h_{V_1}\right) + \left(\alpha^{\Phi}_{V_1P_3} W h_{P_3}\right)\right) =$$

$$= \sigma\big((1 * 0.5 * 0.2) + (1 * 0.5 * 0.4)\big) = 0.3$$

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고칠 점: 해설의 PAP 특징 0.015 는 0.15 의 오타. α_P2P3 칸의 0.125 도 나누기 전 값 (나누면 0.5)

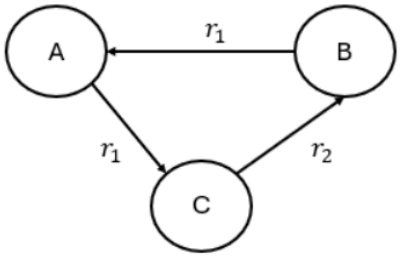
2025 가을 기말 8번 Relational GCN: 관계마다 다른 가중치를 왜 쓰나 + 계산 (b) 와 관련 (관계마다 따로 다룬다)

문제

8. (Relational GCN)(10p) Consider a directed heterogeneous graph. Given $h_A^{(0)} = 1, h_B^{(0)} = 2, h_C^{(0)} = 3$. Given $W_{r_1} = 1, W_{r_2} = 2, W_0 = 1$

Relational GCN: $h_v^{(1)} = \sigma(\sum_{r \epsilon R} \sum_{u \epsilon N_r(v)} \frac{1}{c_{v,r}} W_r h_u^{(0)} + W_0 h_v^{(0)})$

where:
$N_r(v)$ is set of incoming neighbours of node v under relation r,
$c_{v,r} = |N_r(v)|$: relation,
W_r: relation-specific weight,
W_0: self-loop weight.

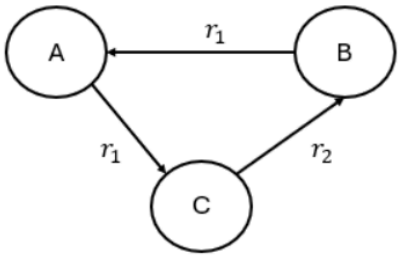


a) Why Relational GCN use different neural network weights for different relation types?
b) Compute $h_A^{(1)}, h_B^{(1)}, h_C^{(1)}$

8. (Relational GCN)(10p) Consider a directed heterogeneous graph. Given $h_A^{(0)} = 1, h_B^{(0)} = 2, h_C^{(0)} = 3$. Given $W_{r_1} = 1, W_{r_2} = 2, W_0 = 1$

Relational GCN: $h_v^{(1)} = \sigma(\sum_{r \epsilon R} \sum_{u \epsilon N_r(v)} \frac{1}{c_{v,r}} W_r h_u^{(0)} + W_0 h_v^{(0)})$

where:
$N_r(v)$ is set of incoming neighbours of node v under relation r,
$c_{v,r} = |N_r(v)|$: relation,
W_r: relation-specific weight,
W_0: self-loop weight.



a) Why Relational GCN use different edges weights for different relation types?
b) Compute $h_A^{(1)}, h_B^{(1)}, h_C^{(1)}$

SOLUTIONS:

a) $h_A^{(1)} = \left(\frac{1}{c_{A,r_1}} W_{r_1} h_B^{(0)} + W_0 h_A^{(0)}\right) = \left(\frac{1}{1} * 1 * 2 + 1 * 1\right) = 9$

$h_B^{(1)} = \left(\frac{1}{c_{B,r_2}} W_{r_2} h_C^{(0)} + W_0 h_B^{(0)}\right) = \left(\frac{1}{1} * 2 * 3 + 1 * 2\right) = 8$

$h_C^{(1)} = \left(\frac{1}{c_{C,r_1}} W_{r_1} h_A^{(0)} + W_0 h_C^{(0)}\right) = \left(\frac{1}{1} * 1 * 1 + 1 * 3\right) = 4$

b) Relational GCN uses different weights per relation because in a heterogeneous graph / knowledge graph, edges have different semantics (different relation types), so messages arriving through different relations should be transformed differently. Using a single shared weight (GCN) would mix fundamentally different relations and reduce the model's ability to represent multi-relational structure.

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고칠 점: 해설의 h_A(1) = 9 는 틀림, 3 이 맞음. 해설의 a), b) 이름표가 문제와 뒤바뀌었고 σ 를 빼먹음